

PÓLYA’S INEQUALITY FOR A SQUARE WITH AN ARBITRARY SQUARE HOLE

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ABSTRACT. We prove Pólya’s inequality for every Dirichlet eigenvalue of a Euclidean square with a closed square removed. The hole is contained in the closed outer square, and the remaining area is positive. Its feasible size, orientation and position are arbitrary; boundary contact and disconnected complements are allowed. Complementary square counts give position-independent spectral comparisons, while restrictions of low square modes and transport between actual form domains control the finite spectral range. For large holes, a mixed-boundary decomposition into sectors and corners retains the deficit needed at the true area. Conformal models for the corner caps preserve Dirichlet energy; their mass weights, endpoint conditions and area costs are explicitly accounted for. Bessel phase bounds and exact lattice-deficit comparisons cover the unbounded ranges, including simultaneous small-angle and small-gap degeneration. Integer and rational certificates handle closed real parameter intervals, not sampled geometries or approximate eigenvalues. We also establish the exact total-side threshold for a universal square-deficit subtraction criterion, yielding a multiple-hole conclusion, and an independent result for large parallel rectangular holes with unrestricted displacement.

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1. INTRODUCTION

The Dirichlet eigenvalues of a planar domain describe the frequencies of a membrane fixed along its boundary. Weyl's law determines their leading density, but does not decide whether the counting function lies below its leading term at every energy. Pólya's conjecture asks for precisely this one-sided statement. For a bounded planar domain Ω , it predicts

$$|\Omega| \lambda_k^D(\Omega) \geq 4\pi k \quad (k \in \mathbb{N}), \quad (1.1)$$

where $\mathbb{N} = \{1, 2, \dots\}$ and eigenvalues are repeated according to multiplicity. The coefficient is fixed by Weyl's law. The challenge is to keep that exact coefficient through the low spectrum, spectral clusters, and the entire unbounded energy range.

Perforated domains make the difficulty particularly transparent. Removing a hole increases each Dirichlet eigenvalue, but decreases the area multiplying it in (1.1). Domain monotonicity alone controls only the first change. Rotation and displacement leave the isolated spectrum of the hole unchanged while changing the geometry of the remaining domain. Near boundary contact, some geometric scales may tend to zero while others remain substantial. An estimate uniform in these configurations cannot simply use a remainder whose constant depends on a positive lower boundary gap.

The complete square-hole family. Let Q be an open square in the Euclidean plane and let H be a closed square of positive side length contained in $\overline{Q}$, with $|Q \setminus H| > 0$. The principal result of this paper is

$$\boxed{|Q \setminus H| \lambda_k^D(Q \setminus H) \geq 4\pi k \quad \text{for every } k \in \mathbb{N}.} \quad (1.2)$$

The side length, orientation and position of H are arbitrary subject only to these conditions. In particular, the assertion includes arbitrary small positive boundary clearances and boundary contact itself; the remaining domain need not be connected. The main work treats strictly interior holes uniformly down to arbitrarily small gaps. A final one-sided approximation then includes contact configurations using only domain monotonicity and convergence of the areas, not an assumed continuity theorem for eigenvalues.

A similarity reduces the outer side length to one. Square symmetries then reduce the relative angle to $0 \leq \theta \leq \pi/4$. If the hole has side q , its axis-aligned bounding box has side

$$b = q\kappa, \quad \kappa = \cos \theta + \sin \theta, \quad 0 < q < 1, \quad b \leq 1. \quad (1.3)$$

Every placement for which $H \subset \overline{Q}$ is allowed. The remaining area is $1 - q^2$, irrespective of angle or displacement. The inequality is invariant under similarities, but not under an arbitrary affine change of coordinates: the latter changes the Euclidean Laplacian. All comparisons below retain the actual Euclidean energy or display explicitly the form to which it is compared.

The inclusive count $N_\Omega(E) = \#\{k : \lambda_k^D(\Omega) \leq E\}$ gives the equivalent formulation

$$N_{Q \setminus H}(E) \leq \frac{|Q \setminus H|}{4\pi} E \quad (E \geq 0). \quad (1.4)$$

Multiplicity and endpoint conventions matter in passing between these two formulations. Our finite-index arguments retain a strict outer square count and an inclusive hole count. Their conclusions are used as indexed eigenvalue bounds, without incorrectly turning an equality at a spectral endpoint into an empty eigenspace.

Two complementary geometric conclusions. There is also a result for finitely many closed square holes H_i contained in the closure of an outer square of side L , with pairwise disjoint interiors. Outer-boundary and mutual contact are allowed. Let their side lengths be l_i , with independent positions and orientations. Put

$$s_* = \sqrt{1 - \frac{12}{5\pi}} = 0.485856226839692\dots \quad (1.5)$$

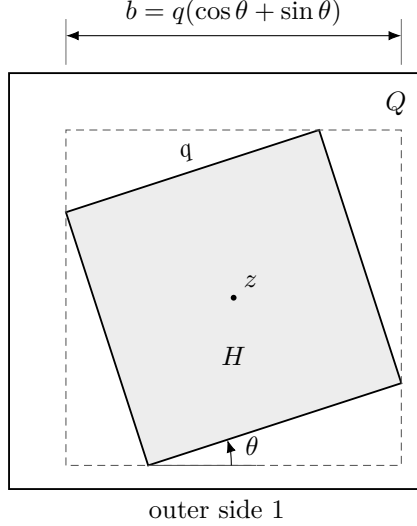


FIGURE 1. A displaced and rotated square hole. The dashed square is its axis-parallel bounding box, not a boundary of the Dirichlet domain. The illustrated configuration has positive clearance; the theorem also permits boundary contact. Solid boundaries carry Dirichlet conditions.

Then

$$\sum_i l_i \leq s_* L \implies \left| Q \setminus \bigcup_i H_i \right| \lambda_k^D \left(Q \setminus \bigcup_i H_i \right) \geq 4\pi k \quad (k \in \mathbb{N}). \quad (1.6)$$

The hypothesis is a sum of side lengths, not a sum of areas. The sharpness assertion concerns its scalar comparison, not an optimal restriction on the geometry. More precisely, with

$$C(t) = \#\{(m, n) \in \mathbb{N}^2 : m^2 + n^2 \leq t^2\}, \quad d(t) = \frac{\pi}{4} t^2 - C(t), \quad (1.7)$$

we prove

$$\sum_i d(q_i t) \leq d(t) \quad \left(q_i \geq 0, \sum_i q_i \leq s_*, t \geq 0 \right), \quad (1.8)$$

and show that no larger total-side threshold guarantees this inequality for all finite lists. The complete one-hole conclusion (1.2) goes beyond this scalar criterion.

Finally, let a single parallel rectangular hole have side lengths pL, qL and arbitrary strictly interior position. We prove

$$\frac{11}{12} \leq p, q < 1 \implies |Q \setminus H| \lambda_k^D(Q \setminus H) \geq 4\pi k \quad (k \in \mathbb{N}). \quad (1.9)$$

The two side ratios and the individual boundary gaps vary independently. This conclusion is not obtained by stretching (1.2); its proof uses a separate two-scale mixed-boundary comparison. The sufficient constant $11/12$ is not asserted optimal.

Prior work and the contribution here. Pólya proved the conjectured inequality for tiling planar domains [1]. The inequalities of Berezin and Li–Yau control eigenvalue averages [2, 3]; these do not directly give the conjectured coefficient at each individual index. Product constructions, unions and packing arguments supply further classes and reductions [4, 5, 6, 7]. Filonov, Levitin, Polterovich and Sher proved Pólya’s conjecture for Euclidean balls [8] and its Dirichlet form for planar circular annuli [9]. The annulus proof combines variational estimates, Bessel analysis, lattice counting and rigorous finite computation. Neither topological equivalence nor conformal equivalence transfers that spectral conclusion to the polygonal domains studied here: conformal change preserves planar Dirichlet energy, but changes the mass in the Rayleigh quotient.

Jiang and Lin develop complementary counting comparisons for perforated strip-tiling domains, as well as explicit estimates with an ϵ -loss for general Lipschitz domains [10]. For square holes in a square, their Theorem 1.15 gives the sufficient total-side condition $\sum_i l_i/L \leq 1/(2\sqrt{2}\pi)$. The variational subtraction principle is an established input here, not a new abstract monotonicity theorem. Our multiple-hole contribution is the exact threshold (1.5) for the stated universal square-deficit comparison, including its finite-list aggregation.

The principal geometric contribution is the full family (1.2). Its proof must keep three types of information together. Complementary square spectra provide position-independent arithmetic bounds. Low-mode restriction and actual-domain transport retain geometric information where those bounds are insufficient. In the near-filling regime, the true-area deficit from four mixed sectors pays for the corner comparison costs. The circular cap models used in this last step have explicitly tracked mass weights and singular endpoint domains; their counts are not substituted for the original count without a form comparison. Horsley's Bessel derivative-phase estimates [14] supply a global radial input, with the first-eigenvalue indexing handled explicitly below.

The disk estimates used in the arithmetic part are elementary Poisson–Bessel bounds with constants chosen for the finite verification. They are not claims to a new Gauss-circle discrepancy record. Likewise, the computer-assisted steps do not calculate approximate eigenvalues of a perforated square. They certify integer and rational inequalities on closed parameter intervals, after analysis has bounded the remaining spectral range. Classical Weyl asymptotics and more recent remainder and Riesz-mean results provide important context [11, 10, 12], but are not used as a replacement for an all-energy proof with the exact coefficient.

How the argument fits together. We first fix the variational and counting conventions and prove the scalar aggregation for small holes. The next part develops the low-mode and transport comparisons for rotated holes. Elementary one-dimensional sums then support the mixed-sector and corner analysis for large holes. The arithmetic section proves the explicit disk estimate, its paired-deficit refinements and the finite-index cover, bringing these arguments together for every strictly interior square hole. One-sided approximation then includes boundary contact. The independent rectangular-hole comparison follows. The appendices specify the additional finite certificates for the multiple-hole and rectangular conclusions.

2. THE DIRICHLET FORM AND COMPLEMENTARY SQUARE COUNTS

We begin with strictly interior, pairwise disjoint holes. These domains are bounded and polygonal, with finitely many separated boundary components. Their Dirichlet Laplacian is the selfadjoint operator associated with

$$\mathfrak{a}_\Omega[u] = \int_\Omega |\nabla u|^2 dx, \quad u \in H_0^1(\Omega),$$

in $L^2(\Omega)$. The resolvent is compact, and

$$\lambda_k^D(\Omega) = \inf_{\substack{V \subset H_0^1(\Omega) \\ \dim V = k}} \sup_{0 \neq u \in V} \frac{\int_\Omega |\nabla u|^2}{\int_\Omega |u|^2}. \quad (2.1)$$

We use the inclusive counting function

$$N_\Omega(E) = \#\{k : \lambda_k^D(\Omega) \leq E\}. \quad (2.2)$$

Thus $N_\Omega(E) \leq |\Omega|E/(4\pi)$ at every energy implies (1.1) by evaluation at an eigenvalue. When a spectral gap gives only a lower bound on an indexed eigenvalue, we use that bound directly rather than infer an inclusive count at the gap endpoint.

Inserting the hole boundaries. Normalize the outer side to one and set

$$t = \frac{\sqrt{E}}{\pi}, \quad e = t^2. \quad (2.3)$$

The eigenvalues of an isolated square of side q are $\pi^2(m^2 + n^2)/q^2$, regardless of its position or orientation. Let P be the outer square with finitely many square holes removed. Extension by zero embeds the direct sum of the Dirichlet form domains on P and the hole interiors into the form domain on the outer square, preserving norms and energies. Min-max gives

$$N_P(E) + \sum_i C(q_i t) \leq C(t). \quad (2.4)$$

In particular, (1.8) implies

$$N_P(E) \leq C(t) - \sum_i C(q_i t) \leq \frac{\pi}{4} \left(1 - \sum_i q_i^2\right) t^2 = \frac{|P|}{4\pi} E. \quad (2.5)$$

The inclusion of form domains, rather than an assumption on eigenfunctions at the interfaces, fixes the direction of this inequality.

This comparison is deliberately insensitive to the arrangement of the holes. It gives the multiple-hole result once the scalar deficits can be aggregated, but does not by itself settle a single large hole. The later geometric comparisons retain information that the two isolated square spectra do not contain. Mixed boundary conditions will occur only on explicitly described comparison pieces; the operator in the main conclusion remains the Euclidean Dirichlet Laplacian.

3. SMALL HOLES AND THE EXACT TOTAL-SIDE COMPARISON

The argument for arbitrarily placed holes is entirely in terms of the deficit d in (1.7). We first construct an envelope that can aggregate any number of holes, and then retain additional partition information where that envelope is too coarse.

A deficit envelope with a monotone slope. The exact reflection identity

$$G(u) = 1 + 4[u] + 4C(u), \quad G(u) = \#\{z \in \mathbb{Z}^2 : |z| \leq u\}, \quad (3.1)$$

and the discrepancy estimate proved in Section 7 give the following slightly weakened convenient bounds:

$$u - \frac{3}{4} - \frac{11}{4}u^{2/3} \leq d(u) \leq u + \frac{1}{4} + \frac{11}{4}u^{2/3} \quad (u \geq 64). \quad (3.2)$$

In particular, for $u \geq 1000$,

$$\frac{2897}{4000} \leq \frac{d(u)}{u} \leq \frac{5101}{4000} < \frac{4}{3}. \quad (3.3)$$

The finite integer checks in Appendix A complete the two bounds

$$d(u) \leq \frac{4}{3}u \quad (u \geq 0), \quad d(u) \leq \frac{1309}{1000}u \quad (0 \leq u \leq \sqrt{18}). \quad (3.4)$$

These upper bounds include left limits at every lattice jump.

Define $F(0) = 0$ and

$$F(u) = \begin{cases} \pi u^2/4, & 0 < u < \sqrt{2}, \\ \max\{\pi\sqrt{2}u/4, \pi u^2/4 - 1\}, & \sqrt{2} \leq u < \sqrt{5}, \\ (1309/1000)u, & \sqrt{5} \leq u < \sqrt{18}, \\ (4/3)u, & u \geq \sqrt{18}. \end{cases} \quad (3.5)$$

The inequality $d(u) \leq F(u)$ follows from the vanishing of C below $\sqrt{2}$, the bound $C \geq 1$ on the next range, and (3.4). Moreover, $F(u)/u$ is nonnegative and nondecreasing. In the second range it is the maximum of a constant and the increasing function $\pi u/4 - 1/u$. At $\sqrt{2}$ there is no downward jump. At $\sqrt{5}$, the left-limit prefix check in (3.4) gives $(5\pi/4 - 1)/\sqrt{5} \leq 1309/1000$, and $\pi\sqrt{2}/4 < 1309/1000$. At $\sqrt{18}$ the slope increases. It follows also that F is nondecreasing.

If $U = \sum_i u_i > 0$, this monotone slope gives

$$\sum_i d(u_i) \leq \sum_i F(u_i) \leq \sum_i u_i \frac{F(U)}{U} = F(U). \quad (3.6)$$

Zero summands and $U = 0$ cause no difficulty. Consequently $\sum_i d(q_i t) \leq F(s_* t)$ under the hypothesis in (1.8).

All but one energy band. Put $R = 243/500$. Rational enclosures of π imply

$$\frac{97}{200} < s_* < R, \quad \frac{\pi}{4}(1 - s_*^2) = \frac{3}{5}. \quad (3.7)$$

The squared transition energies $2/s_*^2, 5/s_*^2, 18/s_*^2$ belong respectively to $(8, 9), (21, 22), (76, 77)$. Write $C_n = C(\sqrt{n})$. For the first quadratic branch of (3.5), the required inequality is $3t^2/5 \geq C(t)$; for the second quadratic branch it is $3t^2/5 \geq C(t) - 1$. The exact tests

$$3n \geq 5C_n \quad (0 \leq n \leq 8), \quad 3n \geq 5(C_n - 1) \quad (8 \leq n \leq 21) \quad (3.8)$$

establish them. The linear branch in the maximum is covered by

$$L_n := 333n - 424C_n \geq 0, \quad (500L_n)^2 \geq 2n(243 \cdot 333)^2 \quad (8 \leq n \leq 21). \quad (3.9)$$

Indeed, division by $\pi/4$ reduces its inequality at $t = \sqrt{n}$ to $n - 4C_n/\pi \geq s_*\sqrt{2n}$. The left side is at least $L_n/333$, and (3.9) proves the stronger bound with R in place of s_* . All quantities are checked nonnegative before squaring.

These are interval checks. On $n \leq t^2 < n + 1$, the count is C_n ; the quadratic differences increase, and the linear difference has derivative $\pi t/2 - \pi\sqrt{2}s_*/4 > 0$ for $t \geq \sqrt{8}$. The overlap at $n = 8$ covers the first noninteger transition.

For the two larger branches, the exact checks give

$$\frac{d(t)}{t} \geq \frac{318087}{500000} = \frac{1309}{1000}R, \quad \sqrt{21} \leq t < \sqrt{77}, \quad t^2 \notin [53, 54), \quad (3.10)$$

$$\frac{d(t)}{t} \geq \frac{81}{125} = \frac{4}{3}R, \quad t \geq \sqrt{76}. \quad (3.11)$$

The finite tests proving these statements are listed in Appendix A. Between successive integer squared radii, $d(t)/t = \pi t/4 - C_n/t$ increases. The infinite part of (3.11) follows from (3.3), since $2897/4000 > 81/125$. The stated ranges overlap the transitions of F , so they prove $d(t) \geq F(s_* t)$ except possibly when $53 \leq t^2 < 54$.

Retaining the partition at the exceptional band. For every finite list $u_i \geq 0$ satisfying $\sum_i u_i \leq B := 18/5$, we have

$$\sum_i d(u_i) \leq \frac{9}{2}. \quad (3.12)$$

Here is the full argument. If every $u_i < \sqrt{2}$, then

$$\sum_i d(u_i) = \frac{\pi}{4} \sum_i u_i^2 \leq \frac{\pi\sqrt{2}}{4} B < \frac{1917}{452} < \frac{9}{2},$$

using $\pi < 355/113$ and $\sqrt{2} < 3/2$. Otherwise retain one actual member $v \geq \sqrt{2}$. Dropping the other counts and summing their quadratic costs gives

$$\sum_i d(u_i) \leq \frac{\pi}{4} \{v^2 + (B - v)^2\} - C(v), \quad \sqrt{2} \leq v \leq B. \quad (3.13)$$

This does not suppose that only one member exceeds $\sqrt{2}$. On the constant-count intervals ending at $\sqrt{5}, \sqrt{8}, \sqrt{10}, B$, the expression is convex. The counts are respectively 1, 3, 4, 6, since $B^2 < 13$. At an interior jump the value drops. It is therefore enough to check the initial endpoint, the three subsequent left limits, and B .

Using the positive lower radical bounds $\sqrt{2} > 7/5$, $\sqrt{5} > 223/100$, $\sqrt{8} > 14/5$, $\sqrt{10} > 79/25$, and $\pi < 355/113$, the corresponding upper bounds for (3.13) are

$$\frac{2488}{565}, \quad \frac{49973}{11300}, \quad \frac{442}{113}, \quad \frac{11349}{2825}, \quad \frac{2361}{565}. \quad (3.14)$$

All are below $9/2$. The radical bounds are verified by squaring, and each endpoint comparison is rational. This proves (3.12).

For $53 \leq t^2 < 54$ the total scaled side satisfies $\sum_i q_i t \leq s_* t < R\sqrt{54} < 18/5$. Meanwhile $C(t) = 37$, and

$$d(t) \geq \frac{53\pi}{4} - 37 > \frac{53}{4} \frac{333}{106} - 37 = \frac{37}{8} > \frac{9}{2}. \quad (3.15)$$

Thus (3.12) closes the entire exceptional band. Together with (3.6) and the preceding comparisons, it proves (1.8) for every real $t \geq 0$. Formula (2.5) now proves the strictly interior, separated-hole case of (1.6); Section 10 includes contact configurations.

The exact limitation of this scalar criterion. If $S > s_*$, choose $s_* < q < \min\{S, \sqrt{2/5}\}$. This is possible because $s_* < \sqrt{2/5}$. At $t = \sqrt{5}$, the inclusive count is $C(t) = 3$, while $C(qt) = 0$. The inequality $d(t) \geq d(qt)$ would require $(5\pi/4)(1 - q^2) \geq 3$, or $q \leq s_*$, a contradiction. Thus no larger threshold on the total side can guarantee (1.8). The comparison itself, rather than the geometry of the perforated domain, is what is sharp here.

For the single-hole argument below it is enough to retain the consequence $q \leq 12/25$, since $12/25 < s_*$. The stronger finite-family conclusion is recorded because its aggregation and sharp scalar threshold do not follow from the single-hole result.

A corridor for moderately sized holes. The finite arithmetic in Appendix A, joined to the explicit discrepancy estimate of Section 7, gives

$$\frac{8}{9}u \leq d(u) \leq \frac{10}{9}u \quad (u \geq 201). \quad (3.16)$$

Here is the exact infinite-range handoff. The weakened bounds (3.2) imply, for $u \geq 15625 = 25^3$,

$$\frac{d(u)}{u} \geq \frac{27811}{31250} > \frac{8}{9}, \quad \frac{d(u)}{u} \leq \frac{17344}{15625} < \frac{10}{9}.$$

The finite part counts all positive lattice pairs through squared radius 15625^2 , retaining inclusive lower values and the left limits needed for upper bounds at jumps. Thus (3.16) holds at every real radius in its stated range.

For $q \leq 4/5$ and $t \geq 201/q$, both radii belong to this corridor, and

$$d(qt) \leq \frac{10}{9}qt \leq \frac{8}{9}t \leq d(t).$$

The complementary count (2.4) therefore proves

$$N_{Q \setminus H}(E) \leq \frac{1 - q^2}{4\pi} E \quad \left(q \leq \frac{4}{5}, E \geq \pi^2(201/q)^2 \right) \quad (3.17)$$

for every feasible orientation and position of the square hole. The finite-index argument can use this uniform onset without any approximation of the perforated-domain spectrum.

4. REFERENCE DOMAINS, ROTATION TRANSPORT, AND THE FINITE LOW MODES

Throughout this section $Q = (0, 1)^2$, H is a closed square strictly contained in Q , and $\Omega = Q \setminus H$. Write q for the side of H , and reduce its angle by the symmetries of Q to $0 \leq \theta \leq \pi/4$. The quantities

$$\kappa = \cos \theta + \sin \theta, \quad b = \kappa q < 1 \quad (4.1)$$

are the angular factor and the side of the axis-parallel bounding square of H . All four gaps between this bounding square and ∂Q are positive; no lower bound on these gaps will be imposed. Every comparison below concerns the Dirichlet form on the actual perforated domain.

4.1. An endpoint-sensitive reference estimate. It is convenient here to count by squared radius:

$$\begin{aligned}\mathcal{C}(s) &= \#\{(m, n) \in \mathbb{N}^2 : m^2 + n^2 \leq s\}, \\ \mathcal{C}_<(s) &= \#\{(m, n) \in \mathbb{N}^2 : m^2 + n^2 < s\}, \quad \mathbb{N} = \{1, 2, \dots\}.\end{aligned}\tag{4.2}$$

Thus $\mathcal{C}(s) = C(\sqrt{s})$ in the radius notation used for the square counts. Ordered pairs are counted separately. In particular, multiplicities are not removed.

For a reference hole of side r , at any feasible angle and position, put

$$j(r, \tau) = \mathcal{C}_<(\tau) - \mathcal{C}(r^2\tau) + 1, \quad \lambda_{j(r, \tau)}(Q \setminus H_r) \geq \pi^2\tau \quad (\tau > 0).\tag{4.3}$$

The outer count is strict and the inner count inclusive. To verify both endpoints, enlarge H_r concentrically, at the same angle, to side $r' > r$, keeping it strictly inside Q , and set $\tau' = \tau(r/r')^2 < \tau$. Inserting the boundary of $H_{r'}$ as a Dirichlet interface gives

$$N_{Q \setminus H_{r'}}(\pi^2\tau') + \mathcal{C}(r'^2\tau') \leq \mathcal{C}(\tau') \leq \mathcal{C}_<(\tau).$$

Here $r'^2\tau' = r^2\tau$. The integer $j(r, \tau)$ is positive and the display proves $\lambda_{j(r, \tau)}(Q \setminus H_{r'}) > \pi^2\tau'$. As $r' \downarrow r$, these domains increase to $Q \setminus H_r$ and each indexed Dirichlet eigenvalue converges to the corresponding eigenvalue of that domain. Indeed, every compactly supported smooth function in the limiting domain belongs to all sufficiently late domains; approximate a finite-dimensional min-max space by such functions and combine this with domain monotonicity. Passing to the limit proves (4.3), including at multiple square eigenvalues. We use its indexed conclusion directly: it need not give a strict inequality for the inclusive counting function at $\pi^2\tau$.

If $r \leq q$, the concentric reference hole at the same angle is a subset of the target hole. Consequently

$$\lambda_k(\Omega) \geq \lambda_k(Q \setminus H_r).\tag{4.4}$$

This is actual set inclusion. It is not an assertion of nesting when normalized bounding-box gaps, rather than the center, are held fixed. The reference bound (4.3) is uniform in position, so we may choose whichever reference placement the relevant comparison requires.

4.2. Increasing the reference side at a fixed angle. Suppose $q < r$ and $\kappa r < 1$. In each coordinate divide the outer interval into its left gap, the bounding-box interval, and its right gap. Send these to intervals with bounding-box length κr , preserving the ratio of the two gaps in that coordinate. The slopes of the resulting increasing piecewise-affine map are

$$\eta = \frac{1 - \kappa r}{1 - \kappa q} < 1 \quad \text{on the gaps}, \quad \sigma = \frac{r}{q} > 1 \quad \text{on the bounding-box interval}.$$

The product map F is a bi-Lipschitz homeomorphism of Q taking H to a square H_r of side r at the same angle. In particular, the map is a homothety on the entire bounding box, not only on the hole. On its complement let $J = \det DF$. The derivative is diagonal with entries in $\{\eta, \sigma\}$, so

$$J \geq \eta^2, \quad JDF^{-1}DF^{-T} \leq \frac{\sigma}{\eta}I.$$

The corner triangles between a rotated hole and its bounding box have derivative σI and energy matrix I ; they therefore satisfy the same bound. For $u \in H_0^1(\Omega)$, the function $v = u \circ F^{-1}$ belongs to $H_0^1(Q \setminus H_r)$, and change of variables gives

$$\int |v|^2 = \int_{\Omega} J|u|^2, \quad \int |\nabla v|^2 = \int_{\Omega} \nabla u^* (JDF^{-1}DF^{-T}) \nabla u.$$

Thus $R(v) \leq \sigma\eta^{-3}R(u)$, where R is the Rayleigh quotient. Min-max proves

$$\lambda_k(\Omega) \geq \frac{q}{r} \left(\frac{1 - \kappa r}{1 - \kappa q} \right)^3 \lambda_k(Q \setminus H_r).\tag{4.5}$$

For later exact interval tests define, when the denominators are positive,

$$T_K(q, r) = \begin{cases} \frac{q}{r} \left(\frac{1 - Kr}{1 - Kq} \right)^3, & q < r, \\ 1, & q \geq r. \end{cases} \quad (4.6)$$

For $q < r$ the logarithmic derivative of the factor in (4.5) with respect to κ is

$$\frac{3(q - r)}{(1 - \kappa r)(1 - \kappa q)} < 0.$$

Hence $K \geq \kappa$ gives a conservative factor, provided $K \max(q, r) < 1$. Moreover, for fixed $K \geq 1$ the derivative of $q(1 - q^2)/(1 - Kq)^3$ has the sign of

$$1 - 3q^2 + 2Kq \geq (1 - q)(1 + 3q) > 0.$$

It follows that $(1 - q^2)T_K(q, r)$ increases below r and decreases above r . On any closed interval $[l, h]$ its minimum is therefore attained at one of the two endpoints, even when $l < r < h$. This observation will replace sampling in the side-length parameter.

4.3. Increasing the reference side by reducing the angle. The preceding size comparison loses strength if κr approaches one. A different map keeps the bounding square fixed. Suppose $0 < \phi < \theta \leq \pi/4$ and put

$$\beta = \frac{\sin \theta}{\cos \theta + \sin \theta}, \quad \beta' = \frac{\sin \phi}{\cos \phi + \sin \phi}, \quad L = \frac{\beta'}{\beta}, \quad U = \frac{1 - \beta'}{1 - \beta}.$$

Then $0 < L < 1 < U$. The reference square has the same bounding square, the same center, and side $r = b/(\cos \phi + \sin \phi) > q$. Every $q < r < b$ is available.

In coordinates in which the bounding square is $[0, b]^2$, the hole vertices are

$$(b\beta, 0), \quad (b, b\beta), \quad (b(1 - \beta), b), \quad (0, b(1 - \beta)).$$

The four complementary corner triangles are mapped affinely to their reference triangles, fixing the bounding-box corners and matching the hole vertices. Their derivative matrices are diagonal, with stretches L, U in one order or the other. The trace on each side of the bounding square is increasing, piecewise affine, and fixes both endpoints. Partition Q outside the bounding square into its four side strips and four exterior corner rectangles. On each side strip apply this trace map in the tangential coordinate and the identity in the normal coordinate; keep each exterior corner rectangle fixed. These maps agree on their interfaces. They may be completed inside the hole by the similarity matching its vertices. This constructs a piecewise-affine bi-Lipschitz homeomorphism of Q taking the target hole to the reference hole, with no assumption on the relative sizes of the four exterior gaps.

On the source perforated domain the Jacobian is LU on an interior corner triangle, L or U on a strip piece, and 1 on an exterior corner rectangle. Thus $J \geq L$. The energy eigenvalues are $U/L, L/U$ on a triangle, $L, 1/L$ or $U, 1/U$ on a strip, and 1, 1 on an exterior corner rectangle. They are all at most U/L . The same change of variables and min-max argument therefore gives

$$\lambda_k(\Omega) \geq \frac{L^2}{U} \lambda_k(Q \setminus H_r). \quad (4.7)$$

Unlike (4.5), this factor has no dependence on the exterior gaps.

For calculation define

$$\beta(x) = \frac{1 - \sqrt{2/x^2 - 1}}{2} \quad (1 \leq x \leq \sqrt{2}), \quad f(z) = \frac{z^2}{1 - z}. \quad (4.8)$$

Both functions are increasing on the ranges used here. Since the reference angular factor is $\kappa' = \kappa q/r$, formula (4.7) becomes

$$\lambda_k(\Omega) \geq \frac{f(\beta(\kappa q/r))}{f(\beta(\kappa))} \lambda_k(Q \setminus H_r). \quad (4.9)$$

The nondegenerate case requires $q < r < \kappa q$; $q = r$ is the identity limit. We never use a collapsed reference angle $\phi = 0$ as a bi-Lipschitz comparison. Nor will we assume that the ratio in (4.9) is monotone in κ : its numerator and denominator are enclosed separately.

4.4. A common core and a hole-mass comparison. There are two useful low-mode estimates that avoid specifying a reference hole. First, for $q > 1/2$ every allowed hole contains the fixed central square

$$[1 - q, q]^2 \subset H. \quad (4.10)$$

Indeed, H contains its concentric axis-parallel square of side q/κ , and either center coordinate lies in $[q\kappa/2, 1 - q\kappa/2]$. The common overlap of these inscribed intervals contains $[1 - q, q]$, because $\kappa + 1/\kappa \geq 2$. Write $\delta = 1 - q$. Domain monotonicity bounds Ω by the shell $Q \setminus [\delta, 1 - \delta]^2$. Neumann cuts divide this shell into four strips of width δ , with Dirichlet conditions on both long sides, and four δ -squares, with Dirichlet conditions on the two adjacent exterior sides and Neumann conditions on the other two sides. The strip eigenvalues start at π^2/δ^2 , and the corner-square eigenvalues at $\pi^2/(2\delta^2)$. The form-domain inclusion for these cuts proves

$$\lambda_1(\Omega) \geq \frac{\pi^2}{2(1 - q)^2}, \quad \frac{(1 - q^2)\lambda_k(\Omega)}{4\pi} \geq \frac{\pi(1 + q)}{8(1 - q)}. \quad (4.11)$$

The second expression increases with q . In a side interval $[l, h]$ this settles every index $k \leq \lfloor p_-(1 + l)/(8(1 - l)) \rfloor$, where $p_- < \pi$.

The second estimate retains the mass that low outer-square modes must place inside the actual hole. This argument works for an arbitrary hole, not only a square. Extend $u \in H_0^1(Q \setminus H)$ by zero to Q and impose orthogonality to the first $k - 1$ outer-square modes. Let $\alpha = \lambda_k(Q)$, choose $\Lambda > \alpha$, and let $\mathcal{U}$ be the span of the remaining outer modes below Λ . Suppose

$$\int_H |w|^2 \geq c \|w\|_{L^2(Q)}^2 \quad (w \in \mathcal{U}), \quad 0 \leq c \leq 1. \quad (4.12)$$

Write $u = u_0 + v$ for its orthogonal low/high decomposition. Since $u = 0$ on H ,

$$\|u_0\|^2 = \langle u, u_0 \rangle = \langle u, \mathbf{1}_{Q \setminus H} u_0 \rangle \leq \sqrt{1 - c} \|u\| \|u_0\|.$$

Consequently $\|v\|^2 \geq c \|u\|^2$. The energy is at least $\alpha \|u_0\|^2 + \Lambda \|v\|^2$, whence the maximum–minimum principle gives

$$\lambda_k(Q \setminus H) \geq \alpha + (\Lambda - \alpha)c. \quad (4.13)$$

The imposed orthogonality conditions have codimension at most $k - 1$; if their restrictions are dependent, the same conclusion remains valid. All modes below the cutoff are retained, including multiplicities.

For the fifth eigenvalue the first four square levels, in units of π^2 , are 2, 5, 5, 8. The next two are 10, with orthonormal basis

$$e_{13}(x, y) = 2 \sin(\pi x) \sin(3\pi y), \quad e_{31}(x, y) = 2 \sin(3\pi x) \sin(\pi y),$$

and the next level is 13. We shall prove, uniformly in angle and position, that their hole-mass Gram matrix is at least $(9/100)I$ whenever $q \geq 3/5$. Formula (4.13) then gives

$$\lambda_5(\Omega) \geq \frac{1027}{100} \pi^2 > \frac{1117}{109} \pi^2 \quad (q \geq 3/5). \quad (4.14)$$

The exact finite records below conservatively use the latter constant. It can also be obtained directly from $\|v\|^2 \geq \|v\|_{L^2(H)}^2 = \|u_0\|_{L^2(H)}^2 \geq c \|u_0\|^2$, which gives $10 + 3c/(1 + c)$ instead of $10 + 3c$. Thus both the sharper statement and the constant actually used by the certificate follow from the same mass calculation.

4.5. The uniform two-mode mass calculation. Assume $q \geq 3/5$ and set $s = 3/(5\kappa)$. The hole contains the concentric axis-parallel square $I \times J$ of side s , and

$$\frac{3}{5\sqrt{2}} \leq s \leq \frac{3}{5}.$$

Since $b = \kappa q \geq 9/(25s)$, each center coordinate z belongs to $[9/(50s), 1 - 9/(50s)]$. For an interval of length s centered at z , direct integration gives

$$\begin{aligned} A_n(s, z) &= \int_I \sin^2(n\pi x) dx = \frac{s}{2} - \frac{\cos(2n\pi z) \sin(n\pi s)}{2n\pi}, \\ B(s, z) &= \int_I \sin(\pi x) \sin(3\pi x) dx = \frac{\cos(2\pi z) \sin(\pi s)}{2\pi} - \frac{\cos(4\pi z) \sin(2\pi s)}{4\pi}. \end{aligned}$$

Define scalar bounds

$$\begin{aligned} a(s) &= \frac{s}{2} + \frac{\sin(\pi s)}{2\pi} \cos\left(\pi \left(1 - \frac{9}{25s}\right)\right), \\ d(s) &= \frac{s}{2} + \frac{\sin(3\pi s)}{6\pi}, \quad b_0(s) = \frac{\sin(\pi s)}{2\pi} + \frac{|\sin(2\pi s)|}{4\pi}. \end{aligned} \quad (4.15)$$

The center restriction gives $A_1 \geq a$. Since $\sin(3\pi s) < 0$ on this range, $A_3 \geq d$, and $|B| \leq b_0$. Both a and d are positive. The Gram matrix on $I \times J$ has diagonal entries $4A_1(I)A_3(J)$ and $4A_3(I)A_1(J)$, and off-diagonal entry $4B(I)B(J)$. Its least eigenvalue is at least

$$4(a(s)d(s) - b_0(s)^2) \geq \frac{9}{100}. \quad (4.16)$$

The first inequality is simply the bound on the two diagonal entries minus the absolute off-diagonal entry. Integration over the rest of H adds a positive-semidefinite matrix. It remains to justify the last numerical inequality over the entire interval, not only at a finite list of values of s .

Here is a full specification of that short rational verification. Let $P_- = 333/106$, $P_+ = 355/113$, $P = (P_- + P_+)/2$, and $\rho = (P_+ - P_-)/2$. The identity

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239) \quad (4.17)$$

and twelve alternating terms of each arctangent, with the next term as an error bound, prove $P_- < \pi < P_+$. For rational a put $x = Pa$ and use the rational enclosures

$$\begin{aligned} S_{\pm}(a) &= \sum_{j=0}^{12} \frac{(-1)^j x^{2j+1}}{(2j+1)!} \pm \left(\frac{|x|^{26}}{26!} + \rho|a| \right), \\ C_{\pm}(a) &= \sum_{j=0}^{12} \frac{(-1)^j x^{2j}}{(2j)!} \pm \left(\frac{|x|^{25}}{25!} + \rho|a| \right). \end{aligned} \quad (4.18)$$

Taylor's theorem and the Lipschitz bounds for sine and cosine prove that these enclose $\sin(\pi a)$ and $\cos(\pi a)$, respectively. All arguments used satisfy $|x| \leq 3$.

For $n = 424, \dots, 599$, let $l = n/1000$, $u = (n+1)/1000$, $m = (l+u)/2$, and define

$$\begin{aligned} s_- &= S_-(m) - P_+(u-l)/2, & s_+ &= S_+(m) + P_+(u-l)/2, \\ a_- &= l/2 + \frac{s_- C_-(1 - 9/(25u))}{2P_+}, & d_- &= l/2 - \frac{S_+(2-3l)}{6P_-}, \\ z &= \max\{|2l-1|, |2u-1|\}, & b_+ &= \frac{s_+}{2P_-} + \frac{S_+(z)}{4P_-}. \end{aligned}$$

The code checks positivity of the factors used in the products. These are valid one-sided bounds because the cosine argument in a increases and stays in $(0, \pi/2)$, because $d'(s) = (1 + \cos(3\pi s))/2 \geq 0$, and because $|\sin(2\pi s)| = \sin(\pi|2s-1|)$ with $\pi|2s-1| < \pi/2$. On every one of the 176 closed intervals the exact rational test is

$$4(a_- d_- - b_+^2) \geq 9/100. \quad (4.19)$$

Their union is $[53/125, 3/5]$, which contains the required range since $(53/125)^2 < 9/50$. Evaluating these explicitly specified tests gives a uniform lower bound of at least $3733/40000 > 9/100$; the smallest enclosure occurs on $[53/125, 17/40]$. The executable `verify_rotation_gram.py` performs precisely this calculation using arbitrary-precision rational numbers. No domain eigenvalues are approximated.

4.6. Three exact angular repairs. The remaining small-index cases in the finite counts are covered by three reference choices. The table gives the complete side interval, target index, reference side, and squared-radius energy. The last three columns show the strict outer count, inclusive inner count, and hence the index certified by (4.3).

k	$[l, h]$	r	τ	$\mathcal{C}_<(\tau)$	$\mathcal{C}(r^2\tau)$	$j(r, \tau)$
9	$[299/400, 3/4]$	$3/4$	32	19	11	9
11	$[3153/4000, 3163/4000]$	$79057/100000$	40	24	15	10
5	$[643/800, 129/160]$	$80623/100000$	20	11	8	4

In particular, the reference index need not equal the target index; $j(r, \tau) \leq k$ is sufficient. The counts in the table are obtained by direct enumeration of the positive ordered pairs through τ .

For all three rows let

$$p_- = \frac{103993}{33102} < \pi < \frac{104348}{33215} = p_+, \quad K_0 = \frac{111}{100}. \quad (4.20)$$

The narrower π bracket is again checked by the twelve-term Machin calculation in (4.17). In the range $\kappa \leq K_0$, (4.5), (4.4), and the endpoint monotonicity proved above reduce the whole row to the rational test

$$\frac{p_- \tau}{4k} \min_{q \in [l, h]} (1 - q^2) T_{K_0}(q, r) \geq 1. \quad (4.21)$$

All three satisfy $K_0 r < 1$. Exact evaluation of the left-hand side, with downward rounding, gives respectively

$$\frac{1169}{1000}, \quad \frac{1013}{1000}, \quad \frac{1025}{1000}$$

as lower bounds, each greater than one. The rational expression (4.21), together with the displayed parameter table, specifies the unrounded values completely. For $q \geq r$, every angle is covered directly by nesting and the separately checked rational inequality

$$\frac{p_-(1 - h^2)\tau}{4k} \geq 1. \quad (4.22)$$

It remains to cover $q \in [l, h_0]$, $h_0 = \min(h, r)$, and $\kappa \in [K_0, 1/l]$. This rectangular parameter set contains all remaining feasible configurations, since $\kappa q < 1$. It may also contain infeasible points; the inequalities below merely overestimate the parameter range. Every row satisfies $l^2 > 1/2$ and $K_0 l > r$. Thus $1/l < \sqrt{2}$, and the positive-angle reference construction is available throughout the feasible part of this set.

For a closed angular interval $[a, b] \subset [K_0, 1/l]$, enclose the two values of β using rational lower and upper bounds. Monotonicity of β and f yields the sufficient test

$$\mathcal{R}(a, b) = \frac{p_-(1 - h_0^2)\tau}{4k} \frac{f(\beta_-(a/r))}{f(\beta_+(b))} \geq 1. \quad (4.23)$$

This follows from (4.9) for every real q and κ in the box, without an assumption on the monotonicity of the quotient in κ .

The radical enclosures in (4.23) are exact. For $x \geq 0$ rational set $M = 2^{60}$ and

$$n = \text{isqrt}(\lfloor xM^2 \rfloor), \quad \frac{n}{M} \leq \sqrt{x} < \frac{n+1}{M}.$$

Here $\text{isqrt}(N)$ is the largest integer whose square is at most N ; the two displayed square inequalities are checked again using integers. Apply this with $x = 2/y^2 - 1$ and reverse the endpoints in $(1 - \sqrt{x})/2$ to obtain $\beta_-(y), \beta_+(y)$. All tested arguments are checked to lie in $[1, \sqrt{2}]$, and the values of f are evaluated only at strictly positive arguments below one.

The complete angular partition can be reconstructed by the following algorithm, with rational arithmetic throughout:

```

stack = [(K0, 1/1, 0)]; leaves = []
while stack is not empty:
    (a,b,d) = pop(stack)
    R = exact_lower_ratio(a,b)          # the displayed leaf test
    if R >= 1:
        append (a,b,R,d) to leaves
    else:
        require d < 30
        m = (a+b)/2
        push (a,m,d+1) and (m,b,d+1)
sort leaves by their left endpoints
require first left endpoint = K0 and last right endpoint = 1/1
require every right endpoint = the next left endpoint

```

The resulting closed covers and downward-rounded minimum ratios are

k	leaves	maximum depth	lower bound for every leaf ratio
9	22	5	125001959/125000000
11	136	9	500084773/500000000
5	54	7	125027239/125000000

In each case the minimum is greater than one. The first two rows are reproduced by `verify_low_mode_rotation_repairs.py`, the third by `verify_fifth_mode_upper_repair.py`. Both scripts also check (4.21), (4.22), the reference counts, and all hypotheses of the transports. Their saved leaf lists are derived outputs, not assumed spectral data. Together with the operator arguments this proves the desired indexed Pólya inequalities on all three closed size bands, for every feasible angle and every strictly interior position.

4.7. The interface with the exact finite census. The unified arithmetic verification in Section 9 uses only explicit consequences of the operator comparisons just proved. We record this interface, including the few extra reference witnesses that are specific to the lower size band. For integer squared-radius energies write $c_e = \mathcal{C}(e)$. Then

$$j_r(\tau) = c_{\tau-1} - c_{\lfloor r^2 \tau \rfloor} + 1 \quad (\tau \in \mathbb{N}) \quad (4.24)$$

is the exact index in (4.3). This retains the strict outer and inclusive inner endpoints.

For a closed target interval $[l, h] \subset [12/25, 7/10]$, take $S = 2^{40}$ and $K_* = 99/70 > \sqrt{2}$. The coefficient belonging to a reference side r is

$$\mu(l, h; r) = \left\lfloor \frac{Sp_-}{4} \min_{q \in [l, h]} (1 - q^2) T_{K_*}(q, r) \right\rfloor. \quad (4.25)$$

Every use checks $K_* \max(h, r) < 1$. By the endpoint monotonicity in Section 4.2, the two integer inequalities

$$j_r(\tau) \leq k, \quad \mu(l, h; r) \tau \geq Sk \quad (4.26)$$

imply $(1 - q^2) \lambda_k(\Omega) \geq 4\pi k$ for the entire real interval $[l, h]$, uniformly in angle and position. Indeed, $\lambda_k(\Omega) \geq T_{K_*}(q, r) \pi^2 \tau$, while $\mu/S \leq p_-(1 - q^2) T_{K_*}(q, r)/4$. In the higher size bands the default reference is $r = l$, and nesting gives the simpler coefficient

$$\mu(l, h; l) = \left\lfloor \frac{Sp_-(1 - h^2)}{4} \right\rfloor \quad (4.27)$$

with no angular feasibility test on an enlarged hole.

On a lower-band interval the fifth mode is accepted directly from (4.14) whenever

$$l \geq 3/5, \quad p_-(1 - h^2) \frac{1117}{109} \geq 20. \quad (4.28)$$

For intervals above $4/5$ the common-core bound (4.11) gives the further analytic alternative $k \leq \lfloor p_-(1+l)/(8(1-l)) \rfloor$. Section 9 also proves the elementary first-two-index bounds and gives the tail-to-finite-index reduction.

The lower band's 28 additional reference witnesses have the following compact description. Each row is used on the successive closed mesh intervals in the displayed union:

k	union of intervals	mesh width	r	τ	$j_r(\tau)$	intervals
3	$[97/200, 499/1000]$	$1/1000$	$1/2$	8	3	14
5	$[123/200, 31/50]$	$1/2000$	$2481/4000$	13	4	10
24	$[1301/2000, 261/400]$	$1/2000$	$653/1000$	61	23	4

Every witness is checked by (4.24), (4.25), and (4.26); the table is not a list of numerically estimated domain eigenvalues. For the two next size bands the only residuals from the default references are exactly the side/index bands proved in Section 4.6. The arithmetic verification reconstructs their missing-index lists and checks inclusion of each whole closed interval in its applicable repair band. Thus the finite records can call these continuum estimates without an angular grid or a placement grid.

4.8. The parallel-angle endpoint needs no separate theorem. The finite comparisons above already include $\kappa = 1$ wherever they are used. For any subsequent argument established only at positive angles, the parallel endpoint follows without invoking a separate parallel-hole theorem. Fix a strictly interior axis-parallel square H_0 of side $q < 1$ and center z . Choose a smooth function χ compactly supported in Q and equal to one in a neighborhood containing H_0 and all its sufficiently small rotations. Let J_0 denote rotation through $\pi/2$, and take the flow of the smooth vector field $\chi(x)J_0(x-z)$. For sufficiently small θ its time- θ map F_θ is the rigid rotation on the hole, the identity near ∂Q , and a diffeomorphism of Q with $DF_\theta \rightarrow I$ uniformly. It takes H_0 to the square H_θ at angle θ , still strictly interior. The pullback mass and energy matrices both converge uniformly to the identity. Rayleigh quotients in the two domains are consequently comparable by factors tending to one, uniformly over nonzero form-domain functions. Min-max gives

$$\lambda_k(Q \setminus H_\theta) \longrightarrow \lambda_k(Q \setminus H_0) \quad (\theta \downarrow 0)$$

for each fixed k . Since the area is always $1 - q^2$, the indexed Pólya inequality passes to this limit. Positive clearance is needed only for each fixed configuration, not uniformly across the family.

5. SQUARE-ROOT SUMS AND MIXED-BOUNDARY COUNTS

The estimates for thin gaps use one-dimensional frequencies. For $x \geq 0$ and $h > 0$, write

$$A_h(x) = \sum_{\substack{j \geq 0 \\ j+1/2 \leq hx}} \sqrt{x^2 - \frac{(j+1/2)^2}{h^2}}, \quad R_h(x) = \sum_{\substack{j \geq 1 \\ j \leq hx}} \sqrt{x^2 - \frac{j^2}{h^2}}, \quad R = R_1. \quad (5.1)$$

Set $A_0 = R_0 = 0$. At a frequency threshold the radical is zero. The Neumann zero modes in a spectral count will be retained separately, even at equality.

The integer-point sum. Let $g_x(y) = \sqrt{x^2 - y^2}$ on $[0, x]$, and put $c_0 = \sqrt{2/27}$. For $x \geq 1$, set $j = \lfloor x \rfloor$ and $r = x - j$. Concavity gives the composite trapezoid estimate

$$R(x) \leq \int_0^j g_x(y) dy - \frac{x}{2} + \frac{g_x(j)}{2}.$$

The chord on $[j, x]$ lies below g_x , so its integral is at least $rg_x(j)/2$. Since $g_x(j) = \sqrt{r(2x-r)} \leq \sqrt{2x}\sqrt{r}$, and the maximum of $(1-r)\sqrt{r}$ on $[0, 1]$ is $2/(3\sqrt{3})$,

$$R(x) \leq \frac{\pi}{4}x^2 - \frac{x}{2} + c_0\sqrt{x} \quad (x \geq 1). \quad (5.2)$$

At an integral x the endpoint contribution is zero. We will also use, for every $x \geq 0$, the simpler right-endpoint-sum inequality

$$R(x) \leq \int_0^x g_x(y) dy = \frac{\pi}{4}x^2. \quad (5.3)$$

The midpoint sum, uniformly down to small widths. For $y \geq 1/2$, let w be the largest half-integer not exceeding y and put $r = y - w \in [0, 1)$. Apply the trapezoid inequality on $[1/2, w]$, use the decreasing initial half-interval, and use the chord on $[w, y]$. This gives

$$A_1(y) - \frac{\pi}{4}y^2 \leq \frac{1-r}{2}\sqrt{y^2 - w^2} \leq c_0\sqrt{y}.$$

The argument includes the case of one summand and a summand at threshold. Below $1/2$ the sum vanishes, so this upper bound remains valid. For $y \geq 3$, the inequality $c_0/\sqrt{3} \leq 4/25$ yields a bound linear in y . The remaining closed interval $[1/2, 3]$ is covered by the 123 rational intervals specified in Appendix B. Thus

$$A_1(y) \leq \frac{\pi}{4}y^2 + \frac{4}{25}y \quad (y \geq 0). \quad (5.4)$$

Both bounds will be needed: the asymptotic one is stronger at a wide gap, while the linear one is uniform as its width shrinks. In particular, $A_h(x) = A_1(hx)/h$ gives

$$A_h(x) - \frac{\pi}{4}hx^2 \leq \min \left\{ \frac{4}{25}x, c_0\sqrt{\frac{x}{h}} \right\} \quad (h > 0). \quad (5.5)$$

One must not use the second estimate with an implicitly bounded $h^{-1/2}$ when a displacement is unrestricted.

6. ANNULAR CORRIDORS AND MIXED CORNER CAPS

The remaining thin-hole estimates must retain the area of every piece. Enclosing a small corner in a rectangle would introduce an area error that eventually dominates the available boundary-sized saving. We instead cut the domain into four annular corridors and four corners, and then treat each corner by a rectangle and two conformal cap models. The model areas are not their physical areas; their explicitly bounded differences will be charged against the corridor savings.

Throughout this section the outer square is $Q = (0, 1)^2$, the hole has side q , and

$$\delta = 1 - q, \quad E = \pi^2 t^2, \quad X = \delta t.$$

Every counting function includes its energy endpoint. In particular, $N_U(E)$ denotes the number of eigenvalues at most E , with multiplicity and with the boundary conditions specified for U . The desired counting inequality is

$$N_{Q \setminus H}(\pi^2 t^2) \leq \frac{\pi}{4}(1 - q^2)t^2. \quad (6.1)$$

We shall prove it for all $t \geq 0$ if $q \geq 49/50$, and for $X \geq 8$ if $q \geq 23/25$. We work first with positive angle $0 < \theta \leq \pi/4$. The zero-angle endpoint is supplied by the smooth-domain continuity argument in Section 4.8; none of the positive-angle estimates below assumes the parallel-hole conclusion.

For a mixed piece U with prescribed Dirichlet edges Γ_D , its form is

$$\mathbf{q}_U[u] = \int_U |\nabla u|^2, \quad \mathcal{D}(\mathbf{q}_U) = H_{\Gamma_D}^1(U).$$

Here $H_{\Gamma_D}^1(U)$ is the closed subspace with zero trace on those edges; no trace restriction is imposed on a Neumann edge. Equivalently one closes smooth functions vanishing near the Dirichlet edges, with the natural finite-energy condition at isolated vertices. At the tangent cusp used below we justify this closure explicitly. Restricting a function in $H_0^1(Q \setminus H)$ to the pieces gives a form embedding into their orthogonal direct sum, with energy and mass both

additive. The min-max principle therefore gives an upper bound for the original count by the sum of the mixed-piece counts. All artificial cuts in this construction are Neumann cuts.

6.1. The four annular corridors. Put

$$c = q \cos \theta, \quad s = q \sin \theta, \quad g = 1 - c - s > 0, \quad w = 1 - c = g + s.$$

The positive gaps between the bounding box of the hole and ∂Q are ℓ, r, d, h , where

$$\ell + r = d + h = g.$$

The hole vertices, in counterclockwise order, are

$$A = (\ell + s, d), \quad B = (1 - r, d + s), \quad C = (\ell + c, 1 - h), \quad D = (\ell, d + c).$$

For example, the supporting lines of the bottom outer side and the hole side AB meet at $O = (\ell + s - d \cot \theta, 0)$. In polar coordinates about O , the region

$$a(d) < \rho < b(d), \quad 0 < \varphi < \theta, \quad a(z) = z \csc \theta, \quad b(z) = c + z \cot \theta$$

lies in the bottom corridor. Its inner arc meets A , and its outer arc meets the bottom side at $(1 - r, 0)$. Both radial sides are portions of the actual Dirichlet boundary. The two circular cuts receive Neumann conditions. Applying the same construction after each quarter turn gives four sectors, with starting gaps ℓ, r, d, h and lengths

$$\mathcal{L}(z) = b(z) - a(z) = c - z \tan(\theta/2), \quad |S_z| = \frac{\theta}{2}(c^2 + 2cz \cot \theta - z^2). \quad (6.2)$$

Their interiors are disjoint: each is contained in the side corridor between the corresponding consecutive hole-vertex coordinates. The complement consists exactly of the four corner regions described below.

For completeness, throughout $q \geq 23/25$ feasibility implies

$$\theta < \frac{1}{10}, \quad \theta \leq \frac{7}{6}\delta, \quad \tan \theta \leq \frac{6}{5}\delta, \quad w \leq \frac{53}{50}\delta, \quad c \geq \frac{91}{100}, \quad \mathcal{L}(z) > \frac{91}{100} \quad (0 \leq z \leq g). \quad (6.3)$$

Indeed the cubic Taylor lower bounds give $\cos(1/10) + \sin(1/10) > 25/23$. On $[0, 1/10]$ they also give $\cos \theta + \sin \theta - 1 \geq (237/250)\theta$; multiplication by $q \geq 23/25$ proves the second assertion. Next use $\tan \theta \leq \theta/(1 - \theta^2/2)$ and $w \leq \delta + \theta^2/2$. Finally $\tan(\theta/2) \leq \frac{1}{2} \tan \theta$ gives

$$\mathcal{L}(z) \geq 1 - \frac{53}{50}\delta - \frac{3}{5}\delta^2 > \frac{91}{100}.$$

In particular all four annular sectors have positive inner radius and positive length. Only polynomial budget functions, never a radial operator at zero radius, will be evaluated at the formal endpoints $z = 0, g$.

6.2. The exact corner partition. At the bottom-left corner put

$$L = \ell + s, \quad B_T = L \cot \theta = c + \ell \cot \theta, \quad a = d \csc \theta.$$

Cut along $y = d$ and $x = L$. One piece is the rectangle $(0, L) \times (0, d)$, with Dirichlet conditions on its two outer-square sides and Neumann conditions on the two cuts. The piece above it is a tangent cap T , bounded by the line $y = d$, the circle of radius B_T centered at $(0, d + B_T)$, and the actual hole side DA . The line and circle are Neumann; the hole side is Dirichlet. The piece to its right is a chord cap C_* , bounded by $x = L$, the circle of radius a centered at $(L - d \cot \theta, 0)$, and $y = 0$. The chord and circle are Neumann; $y = 0$ is Dirichlet.

Translating the top tangent contact to the origin, the top cap is

$$x = \rho \sin \varphi, \quad y = B_T - \rho \cos \varphi, \quad 0 < \varphi < \theta, \quad B_T < \rho < B_T \sec \varphi.$$

For the right cap, translating its circle center to zero gives

$$0 < \varphi < \theta, \quad a \cos \theta \sec \varphi < \rho < a.$$

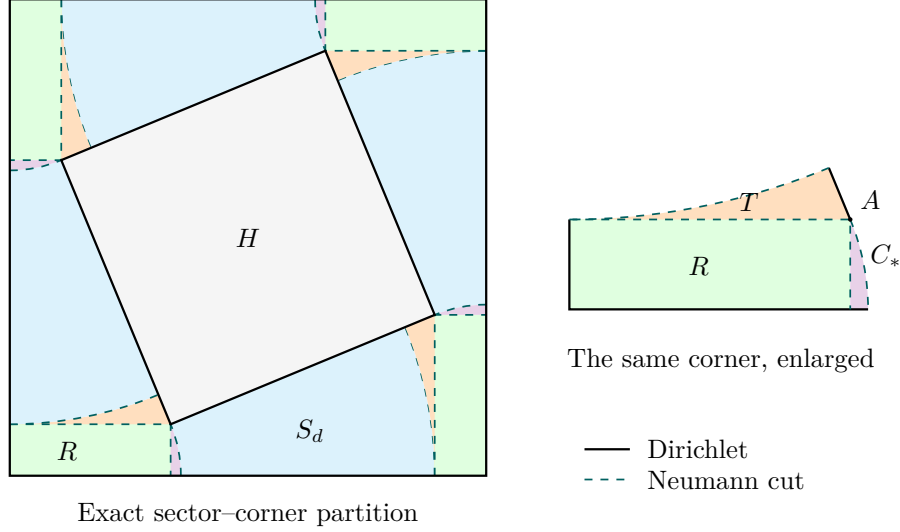


FIGURE 2. The exact partition, illustrated with $q = 3/5$, $\theta = \pi/8$, and equal bounding-box gaps. These larger gaps make the cap geometry visible; the estimates use the thin ranges specified in the text. Each cyan annular corridor has Dirichlet rays and Neumann arcs. Each corner is exactly the union of the rectangle R , tangent cap T , and chord cap C_* ; no area is discarded.

These descriptions verify both the partition and its boundary labels. The top cap is above $y = d$ and left of $x = L$; the right cap is right of $x = L$ and below $y = d$; the rectangle occupies the remaining corner. Their exact areas are

$$|T| = \frac{B_T^2}{2}(\tan \theta - \theta), \quad |C_*| = \frac{d^2}{2 \sin^2 \theta}(\theta - \sin \theta \cos \theta). \quad (6.4)$$

The four corner parameters satisfy

$$\sum L = 2g + 4s \leq 4w, \quad \sum d = 2g, \quad L \leq w, \quad d \leq g, \quad \sum L^2 \leq 4w^2, \quad \sum d^2 \leq 2g^2. \quad (6.5)$$

The last bound follows by pairing opposite gaps: for $x + y = g$, $x^2 + y^2 \leq g^2$.

6.3. A radial Neumann count with its first root included. We first establish the radial input, including its counting convention. For $0 < a < b$ and real $\nu \geq 0$ consider

$$\mathcal{L}_\nu = -\frac{d^2}{dr^2} - \frac{1}{r} \frac{d}{dr} + \frac{\nu^2}{r^2}$$

in $L^2((a, b), r dr)$, with form domain $H^1(a, b)$ and form $\int_a^b (r|u'|^2 + \nu^2|u|^2/r) dr$. Its operator conditions are the ordinary radial conditions $u'(a) = u'(b) = 0$. The endpoints are positive and regular; no condition at a Bessel singularity is involved.

Put

$$M_\nu(x)^2 = J'_\nu(x)^2 + Y'_\nu(x)^2, \quad J'_\nu(x) + iY'_\nu(x) = M_\nu(x)e^{i\phi_\nu(x)},$$

using the continuous phase with limit $\pi/2$ at zero. The Wronskian and Bessel equation give

$$\phi'_\nu(x) = \frac{2(x^2 - \nu^2)}{\pi x^3 M_\nu(x)^2}. \quad (6.6)$$

We use the derivative-modulus inequality of Horsley [14, Theorem 2, equation (15), and Section 4, equation (58)]:

$$\frac{d}{dx} \left(\frac{x^2 M_\nu(x)^2}{\sqrt{x^2 - \nu^2}} \right) < 0, \quad \frac{x^2 M_\nu(x)^2}{\sqrt{x^2 - \nu^2}} > \frac{2}{\pi}, \quad x > \nu \geq 0. \quad (6.7)$$

The second inequality follows from the first and its limiting value $2/\pi$ at infinity. These are global inequalities for every real nonnegative order, not asymptotic estimates with an unspecified onset. Together with (6.6), they imply

$$\phi'_\nu(x) < \frac{\sqrt{x^2 - \nu^2}}{x} \quad (x > \nu), \quad \phi'_\nu(x) < 0 \quad (0 < x < \nu). \quad (6.8)$$

The phase derivative vanishes at the turning point when $\nu > 0$.

For $\sigma > 0$ define

$$F(\sigma) = \phi_\nu(b\sigma) - \phi_\nu(a\sigma), \quad I_\nu(\sigma^2) = \frac{1}{\pi} \int_a^b \sqrt{\sigma^2 - \nu^2/r^2} \, dr.$$

To identify the count, use the solution normalized at the left endpoint,

$$u(r; \sigma) = c_\sigma [Y'_\nu(a\sigma)J_\nu(r\sigma) - J'_\nu(a\sigma)Y_\nu(r\sigma)], \quad c_\sigma = \frac{\pi a \sigma}{2}.$$

The Wronskian $J_\nu Y'_\nu - J'_\nu Y_\nu = 2/(\pi x)$ gives $u(a; \sigma) = 1$, $u_r(a; \sigma) = 0$, and

$$u_r(b; \sigma) = -\sigma c_\sigma M_\nu(a\sigma)M_\nu(b\sigma) \sin F(\sigma).$$

Thus a positive eigenvalue occurs precisely at an integer phase level. At $F(\sigma) = j\pi$, the endpoint derivative vectors are proportional, and the same Wronskian gives

$$b u(b; \sigma) \partial_\sigma u_r(b; \sigma) = -\frac{2}{\pi} c_\sigma^2 M_\nu(a\sigma)^2 F'(\sigma).$$

Differentiating the radial equation in σ gives the Lagrange identity

$$\frac{d}{dr} [r(u \partial_\sigma u_r - u_r \partial_\sigma u)] = -2\sigma r u^2.$$

Its left endpoint term is zero. At a Neumann root it therefore proves

$$F'(\sigma) = \frac{\pi \sigma}{c_\sigma^2 M_\nu(a\sigma)^2} \int_a^b r u(r; \sigma)^2 dr > 0. \quad (6.9)$$

In particular, global monotonicity of F is not being assumed.

If $\nu > 0$, the small-argument expansions and (6.6) show that $-\pi/2 < F(\sigma) < 0$ for sufficiently small positive σ . The large-argument expansions give $F(\sigma) = (b-a)\sigma + o(1)$; the relevant standard function and derivative expansions are recorded in [15, Sections 10.7 and 10.17]. Every integer-level crossing is upwards by (6.9). There can be no negative integer-level crossing, nor two crossings of the same integer level: either would require an intervening contact with nonpositive derivative. The first eigenvalue is consequently the crossing $F = 0$, followed by $F = \pi, 2\pi, \dots$. For $\nu = 0$, the phase is increasing and $F(\sigma) > 0$ tends to zero as $\sigma \downarrow 0$. The positive roots then start at $F = \pi$, and the constant function contributes the zero eigenvalue. Regular Sturm–Liouville uniqueness ensures that these roots are precisely the simple radial eigenvalues. We have thus proved, including every positive energy endpoint,

$$N_\nu^{NN}(\sigma^2) = 1 + \left\lfloor \frac{F(\sigma)}{\pi} \right\rfloor. \quad (6.10)$$

Integrating (6.8) from $a\sigma$ to $b\sigma$ yields $F(\sigma) < \pi I_\nu(\sigma^2)$; strictness holds because the interval has positive length and the derivative inequality is strict except possibly at one point. The forbidden part of the interval only helps. Therefore

$$N_\nu^{NN}(E) < I_\nu(E) + 1, \quad N_\nu^{NN}(E) \leq \lceil I_\nu(E) \rceil, \quad E > 0. \quad (6.11)$$

In particular zero action gives zero count, even at a turning threshold. At $E = 0$ the counts are separately zero for $\nu > 0$ and one for $\nu = 0$. This last distinction avoids applying the strict inequality at zero energy to the constant Neumann mode.

6.4. **The scalar radial deficit.** Write

$$R(y) = \sum_{m \geq 1} \sqrt{y^2 - m^2}_+, \quad D(y) = \frac{\pi y}{4} - \frac{R(y)}{y} \quad (y > 0), \quad D(0) = 0.$$

We need only the following modest uniform lower bound:

$$D(y) \geq \frac{7}{20} \quad (y \geq 1), \quad D(y) = \frac{\pi y}{4} \quad (0 \leq y \leq 1). \quad (6.12)$$

Here is a short analytic tail and a complete finite specification of the small remaining interval. For $y \geq 1$, put $n = \lfloor y \rfloor$, $h = y - n$, and $f(x) = \sqrt{y^2 - x^2}$ on $[0, y]$. Concavity on $[0, n]$ gives

$$\int_0^n f(x) dx \geq \frac{y}{2} + \sum_{m=1}^{n-1} f(m) + \frac{f(n)}{2}.$$

Moreover,

$$\int_n^y f(x) dx = \int_0^h \sqrt{u(2y-u)} du \geq \frac{2}{3} h^{3/2} \sqrt{2y-h}.$$

It follows that

$$\frac{\pi y^2}{4} - R(y) \geq \frac{y}{2} - \sqrt{2y-h} \sqrt{h} \left(\frac{1}{2} - \frac{2h}{3} \right).$$

For $h \geq 3/4$ the correction on the right is nonnegative. For $0 \leq h \leq 3/4$, the maximum of $\sqrt{h}(1/2 - 2h/3)$ is $1/6$, attained at $h = 1/4$. Consequently

$$D(y) \geq \frac{1}{2} - \frac{1}{\sqrt{18y}} > \frac{7}{20}, \quad y \geq \frac{5}{2}, \quad (6.13)$$

where the last comparison is $1/45 < 9/400$.

For $1 \leq y \leq 5/2$, use the 75 closed intervals $[a_j, b_j] = [1 + j/50, 1 + (j+1)/50]$, $0 \leq j < 75$. The monotonicity of each radical gives the rigorous lower bound

$$D(y) \geq \frac{333}{106} \frac{a_j}{4} - \sum_{m=1}^{\lfloor b_j \rfloor} \sqrt{1 - m^2/b_j^2}, \quad a_j \leq y \leq b_j. \quad (6.14)$$

To evaluate it, for every nonnegative rational x let

$$n_x = \left\lfloor \sqrt{\lfloor 2^{100} x \rfloor} \right\rfloor, \quad \sqrt{x} < \frac{n_x + 1}{2^{50}}.$$

Integer squaring verifies both this strict upper bound and $(n_x/2^{50})^2 \leq x$. The inequality $333/106 < \pi$ is checked using $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$ and the first twelve terms of each alternating series, with its next term as remainder bound. Substituting the radical upper bounds in (6.14), exact rational arithmetic gives a lower bound at least

$$\frac{36535721}{100000000} > \frac{7}{20}$$

on every interval. The supplied verifier `verify_radial_scalar.py` performs exactly these 75 tests, checks the adjoining endpoints, and uses integer and rational arithmetic only. Its explicit failure checks are unaffected by Python's `-O` option. Equations (6.13) and (6.14) prove (6.12) for all real y .

An averaging consequence will be important at small inner radii. If $0 \leq u < v$ and $v \geq 1$, then

$$\int_u^v D(y) dy \geq \frac{7}{20} (v - u). \quad (6.15)$$

If $u \geq 1$ this is pointwise. Otherwise the average on $[u, 1]$ is $\pi(1+u)/8 \geq \pi/8 > 7/20$, and the remaining interval is covered by (6.12).

6.5. The true-area corridor saving. On a sector with Dirichlet rays and Neumann arcs the angular orders are $\nu_m = m\pi/\theta$, $m \geq 1$. By (6.11), its count is bounded by the sum of the ceilings of

$$I_m = \int_a^b \sqrt{t^2 - m^2/(\theta\rho)^2} + d\rho.$$

The channels with $m \geq \theta bt$ have zero action and zero count; including equality in $M = \lfloor \theta bt \rfloor$ is harmless. Thus

$$\begin{aligned} N_{S_z}(\pi^2 t^2) &\leq \sum_{m=1}^M \lceil I_m \rceil \\ &\leq \frac{\pi}{4} |S_z| t^2 - t \int_{a(z)}^{b(z)} D(\theta \rho t) d\rho + \lfloor \theta b(z) t \rfloor. \end{aligned} \quad (6.16)$$

This has the true sector area coefficient. If $\theta b(z)t < 1$, the count is zero. Otherwise (6.15) shows that the normalized Weyl saving is at least $\frac{7}{20}\mathcal{L}(z) - \theta b(z)$.

Define the two budgets

$$\begin{aligned} \mathcal{A}(z) &= \frac{7}{20}\mathcal{L}(z) - \theta b(z), \\ \mathcal{B}(z) &= \frac{\pi}{4} |S_z| t = \frac{\pi \theta t}{8} (c^2 + 2cz \cot \theta - z^2), \\ \mathcal{G}(z) &= \min\{\mathcal{A}(z), \mathcal{B}(z)\}. \end{aligned} \quad (6.17)$$

The first budget applies to an active sector; the second is its exact saving if inactive. Hence $t\mathcal{G}(z)$ is always a valid saving. The identity

$$\theta b(g) = \theta \cot \theta w \leq w$$

and (6.3) show that $\mathcal{A}(z) \geq 2337/10000 > 0$; $\mathcal{B}$ is nonnegative as well. The function $\mathcal{A}$ is affine, and $\mathcal{B}$ is concave, so $\mathcal{G}$ is concave: its hypograph is the intersection of their convex hypographs. Consequently

$$\mathcal{G}(z) + \mathcal{G}(g - z) \geq \mathcal{G}(0) + \mathcal{G}(g).$$

The two opposite-gap identities therefore give the uniform placement bound

$$\sum_{z \in \{\ell, r, d, h\}} N_{S_z}(\pi^2 t^2) \leq \frac{\pi}{4} \left(\sum_z |S_z| \right) t^2 - 2t(\mathcal{G}(0) + \mathcal{G}(g)). \quad (6.18)$$

Also $\mathcal{B}$ increases on $[0, g]$, because its derivative has the sign of $c \cot \theta - z > 0$ by (6.3).

6.6. An elementary action estimate for decreasing potentials. The cap models require a different one-dimensional estimate. Let V be nonnegative, smooth and strictly decreasing on $(0, L)$, with $V(0+) = +\infty$, and use the Friedrichs Dirichlet form

$$\int_0^L (|u'|^2 + V|u|^2), \quad \{u \in H_0^1(0, L) : \int_0^L V|u|^2 < \infty\}.$$

For every $E > 0$ its inclusive count satisfies

$$N_V(E) \leq \frac{1}{\pi} \int_0^L \sqrt{E - V} + dx + \mathcal{H}(L\sqrt{E}), \quad \mathcal{H}(z) = \frac{1}{2} + \frac{1}{2\pi} \log_+ z. \quad (6.19)$$

To see this, put $p = \sqrt{E - V} +$. If $p(L) \leq 1/L$, then $V \geq E - L^{-2}$, and the DD kinetic lower bound π^2/L^2 makes the inclusive count zero. Otherwise let x_0 be the unique point where $p(x_0) = 1/L$ and introduce a Neumann cut there. On $(0, x_0)$, the first eigenvalue is at least

$$E - L^{-2} + \frac{\pi^2}{4x_0^2} > E,$$

so this piece contributes no count. This comparison remains valid for the singular left endpoint because its Friedrichs form has zero trace. On the right interval solve $u(x_0) = 1$, $u'(x_0) = 0$ and define

$$u = \rho \sin \vartheta, \quad u' = p \rho \cos \vartheta, \quad \vartheta(x_0) = \pi/2.$$

Differentiation gives

$$\vartheta' = p + \frac{p'}{2p} \sin(2\vartheta) \leq p + \frac{p'}{2p}.$$

At a zero of u , the phase crosses an integer multiple of π upwards, with derivative $p > 0$. Sturm oscillation for the N-D interval gives the inclusive count $\lfloor \vartheta(L)/\pi \rfloor$; an endpoint zero is counted. Integration bounds it by

$$\frac{1}{2} + \frac{1}{\pi} \int_{x_0}^L p \, dx + \frac{1}{2\pi} \log(Lp(L)),$$

which proves (6.19).

6.7. The tangent cap and its Kelvin endpoint. Translate the top contact to zero and invert by

$$(u, v) = \frac{(x, y)}{x^2 + y^2}.$$

The tangent line and circle become the Neumann lines $v = 0$ and $v = h_0 = 1/(2B_T)$. The Dirichlet hole edge $x \cos \theta + y \sin \theta = B_T \sin \theta$ becomes $u = U(v)$, where

$$U(v) = h_0 \cot \theta + \sqrt{h_0^2 \csc^2 \theta - (v - h_0)^2}, \quad 0 < v < h_0.$$

The image is $u > U(v)$, and $\min U = U(0) = \cot \theta / B_T =: U_0$. Zero extension into the product strip $u > U_0$, $0 < v < h_0$ crosses only this Dirichlet crosscut; it does not move a Neumann boundary. Dirichlet energy is invariant under this two-dimensional conformal or anticonformal map. Its exact mass is

$$\int |F(u, v)|^2 (u^2 + v^2)^{-2} \, du \, dv.$$

Replacing the weight by the larger u^{-4} lowers all Rayleigh quotients. The two weights are comparable on the strip because $v/u \leq h_0/U_0$, so the extension maps into the finite-mass model form domain. Its count is consequently an upper bound for the cap count. More explicitly, the model form domain is

$$\mathcal{V} = \left\{ F \in H_{\text{loc}}^1((U_0, \infty) \times (0, h_0)) : F|_{u=U_0} = 0, \quad \int (|\nabla F|^2 + u^{-4}|F|^2) \, du \, dv < \infty \right\}.$$

There is no boundary restriction at infinity or on the two horizontal sides. The Neumann cosine expansion in the transverse variable is an orthogonal decomposition of both its energy and weighted mass.

Separate the transverse NN modes $\mu_m = m\pi/h_0$, $m \geq 0$. For $x = 1/u$ and $F(u) = G(x)/x$ one has $0 < x < L = 1/U_0 = B_T \tan \theta$, and direct substitution gives

$$\begin{aligned} \int_{U_0}^{\infty} u^{-4} |F|^2 \, du &= \int_0^L |G|^2 \, dx, \\ \int_{U_0}^{\infty} |F'|^2 \, du &= \int_0^L |G'|^2 \, dx - \left[\frac{|G|^2}{x} \right]_0^L, \\ \mu_m^2 \int_{U_0}^{\infty} |F|^2 \, du &= \int_0^L \mu_m^2 x^{-4} |G|^2 \, dx. \end{aligned}$$

The resulting operator is exactly the Friedrichs DD realization of

$$-G'' + \mu_m^2 x^{-4} G. \tag{6.20}$$

Here the endpoint at infinity deserves attention. For the zero mode, $F(U_0) = 0$ and $F' \in L^2(U_0, \infty)$ imply $F(u) = o(\sqrt{u})$: split the derivative integral at a fixed large point and apply

Cauchy–Schwarz to its tail. Hence $G(x)^2/x \rightarrow 0$ at zero, and $G(0) = 0$. The half-line Hardy inequality gives $\int_{U_0}^\infty |F|^2/u^2 \leq 4 \int_{U_0}^\infty |F'|^2$, and $G'(x) = F(1/x) - x^{-1}F'(1/x)$ then gives $G' \in L^2(0, L)$. Conversely, closing the compactly supported identity above gives the whole DD form. Cutoffs at strip infinity are dense in the original natural form: truncate the function values first, then let the cutoff interval length tend to infinity. Its added energy tends to zero. Thus a finite-energy constant limit of F is allowed and corresponds to G proportional to x ; no extra physical Dirichlet condition has been imposed at the cusp.

The zero channel of (6.20) has count $\lfloor Lt \rfloor \leq Lt$. A positive channel can contribute only if $m \leq h_0 L^2 t$. Apply (6.19) to those channels, and bound the decreasing sum of their actions by the integral over all nonnegative real orders. This gives

$$\sum_{m \geq 1} \int_0^L \sqrt{t^2 - \frac{m^2}{h_0^2 x^4}} dx \leq \frac{\pi t^2}{4} \int_0^L h_0 x^2 dx.$$

With $W = h_0 L^2 = L \tan \theta/2$ and $A_T^{\text{mod}} = h_0 L^3/3 = B_T^2 \tan^3 \theta/6$, we conclude

$$N_T(\pi^2 t^2) \leq \frac{\pi}{4} A_T^{\text{mod}} t^2 + Lt + \lfloor Wt \rfloor \mathcal{H}(\pi Lt). \quad (6.21)$$

Its leading area excess is not discarded. Writing $z = \tan \theta$,

$$A_T^{\text{mod}} - |T| = \frac{B_T^2}{2} \left(\frac{z^3}{3} - z + \arctan z \right) = \frac{B_T^2}{2} \int_0^z \frac{v^4}{1+v^2} dv \leq \frac{L^2 \tan^3 \theta}{10}. \quad (6.22)$$

6.8. The chord cap and its zero angular channel. Place the right-cap circle center at zero, and put $A_+ = a \cos \theta + id$, $A_- = a \cos \theta - id$. The Möbius map

$$z = -\frac{\zeta - A_+}{\zeta - A_-}$$

takes the upper circular segment exactly onto $0 < |z| < 1$, $0 < \arg z < \theta$. On the chord it equals $(d-y)/(d+y) > 0$, the circular arc has argument θ , and the bottom real segment has modulus one. Thus the two rays are Neumann and the outer arc is Dirichlet. Its pole is outside the half cap, and the map is nondegenerate at the vertex sent to zero. The inverse derivative has squared modulus $4d^2/|1+z|^4$.

For $z = \rho e^{i\varphi}$,

$$|1+z|^2 = (1+\rho)^2 - 4\rho \sin^2(\varphi/2) \geq (1+\rho)^2 \cos^2(\theta/2).$$

Set $\gamma = \sec^4(\theta/2)$ and $t' = \sqrt{\gamma} t$. The actual cap count at E is at most the count at γE of the radial mass model $4d^2/(1+\rho)^4$ on the same wedge. This is a mass comparison on a fixed form domain, not Neumann domain monotonicity. The substitution

$$x = \frac{2d\rho}{1+\rho}, \quad 0 < x < d, \quad f(x) = x \left(1 - \frac{x}{2d} \right)$$

transforms the metric to $dx^2 + f(x)^2 d\varphi^2$. Its model area is

$$A_C^{\text{mod}} = \theta \int_0^d f(x) dx = \frac{\theta d^2}{3}.$$

For angular NN order $\nu_m = m\pi/\theta$, $m \geq 0$, the form and mass are

$$\int_0^d \left(f|u'|^2 + \frac{\nu_m^2}{f} |u|^2 \right) dx, \quad \int_0^d f|u|^2 dx. \quad (6.23)$$

The form domain has trace $u(d) = 0$ and finite displayed energy at zero. This is the regular sector condition; for $m = 0$ it allows a nonzero constant limit, whereas the logarithmic solution has infinite energy.

For $m \geq 1$, $G = \sqrt{f} u$ gives the Friedrichs DD operator

$$-G'' + \frac{\nu_m^2 - 1/4}{f^2} G, \quad 2ff'' - (f')^2 = -1.$$

The transformation is first an identity on compactly supported forms and then by closure. For eigenfunctions the regular behavior $u = O(x^{\nu_m})$ also makes its endpoint term vanish. Put $\beta = \sqrt{1 - \theta^2}/(4\pi^2)$. Since $\nu_m^2 - 1/4 \geq \beta^2 \nu_m^2$, replacing the potential by $\beta^2 \nu_m^2/f^2$ lowers the eigenvalues. This nonnegative potential is strictly decreasing on $(0, d)$, so (6.19) applies at energy $\pi^2 t'^2$.

The number of channel errors is determined before this replacement. The original form (6.23) gives $\lambda \geq \nu_m^2/\max f^2 = 4\nu_m^2/d^2$, so only $m \leq \theta dt'/2$ can contribute. It is safe to enlarge the nonnegative action sum to all $m \geq 1$ while keeping this original cutoff for the error terms. The decreasing-sum integral then bounds the action sum by $\pi A_C^{\text{mod}} t'^2/(4\beta)$.

For $m = 0$ the critical negative potential must not be dropped. Instead let $-(fu')' = \lambda fu$, $\lambda > 0$, and set $v = fu'$, $G = v/\sqrt{f} = \sqrt{f} u'$. Direct differentiation gives

$$-G'' + V_+(x)G = \lambda G, \quad V_+ = \frac{3(f')^2}{4f^2} - \frac{f''}{2f} > 0. \quad (6.24)$$

A regular original eigenfunction satisfies $u(x) = C - \lambda C x^2/4 + O(x^3)$, and hence $G(x) = -\lambda C x^{3/2}/2 + O(x^{5/2})$. The partner potential is $3/(4x^2) + O(1/x)$; its other local solution has exponent $-1/2$ and is not square integrable. Thus zero is the limit-point, Dirichlet-trace endpoint of the partner. At d , $v'(d) = -\lambda f(d)u(d) = 0$ and $f'(d) = 0$, so $G'(d) = 0$. Conversely, any positive partner eigenfunction gives

$$u = -\frac{(\sqrt{f} G)'}{\lambda f}, \quad u' = \frac{G}{\sqrt{f}},$$

which is regular at zero and vanishes at d . These transformations are inverse on positive eigenspaces. The original has no zero mode because zero energy forces a constant with $u(d) = 0$; the partner has none because its form is positive. There is therefore no deleted ground state or exceptional lost eigenvalue.

Since $V_+ \leq C_d/x^2$, Hardy's inequality shows that the partner form domain is $\{G \in H^1(0, d) : G(0) = 0\}$. Its form dominates the ordinary D-N kinetic form, whose eigenvalues are $((j - 1/2)\pi/d)^2$, $j \geq 1$. Consequently its inclusive count, and the original zero-channel count, are at most $\lfloor dt' + 1/2 \rfloor$. Combining all channels gives

$$N_{C_*}(\pi^2 t^2) \leq \frac{\pi \gamma A_C^{\text{mod}}}{4\beta} t^2 + \lfloor dt' + 1/2 \rfloor + \left\lfloor \frac{\theta dt'}{2} \right\rfloor \mathcal{H}(\pi dt'). \quad (6.25)$$

Finally, sine concavity on $[0, \theta]$ gives

$$|C_*| = a^2 \int_0^\theta \sin^2 \varphi d\varphi \geq \frac{a^2 \sin^2 \theta \theta}{3} = A_C^{\text{mod}}.$$

The full excess over the true area, including the energy and angular comparisons, is therefore bounded by

$$\frac{\gamma A_C^{\text{mod}}}{\beta} - |C_*| \leq \frac{\theta d^2}{3} \left(\frac{\gamma}{\beta} - 1 \right). \quad (6.26)$$

6.9. Rectangle counts and two forms of the corner bound. The mixed rectangle eigenvalues are $\pi^2((j + 1/2)^2/L^2 + (k + 1/2)^2/d^2)$, $j, k \geq 0$. Write $A_1(y) = \sum_{j \geq 0} \sqrt{y^2 - (j + 1/2)^2}_+$. The scalar midpoint bound (5.4) is $A_1(y) \leq \pi y^2/4 + 4y/25$. Summing one index and using $\lfloor v + 1/2 \rfloor \leq v + 1/2$ gives

$$N_R(\pi^2 t^2) \leq \frac{\pi}{4} L d t^2 + \left(\frac{4}{25} d + \frac{L}{2} \right) t + \frac{1}{4}. \quad (6.27)$$

Indeed the radical sum is $(d/L)A_1(Lt)$, and the remaining half-count is at most $\frac{1}{2} \lfloor Lt + 1/2 \rfloor \leq Lt/2 + 1/4$. Interchanging L, d gives the other version. Averaging yields excess at most $33(L + d)t/100 + 1/4$. If the rectangle is active, its first eigenvalue implies $(L + d)t \geq \sqrt{2}$. If inactive, its count is zero. Thus at every energy we also have

$$N_R(\pi^2 t^2) \leq \frac{\pi}{4} L d t^2 + \frac{51}{100} (L + d)t, \quad (6.28)$$

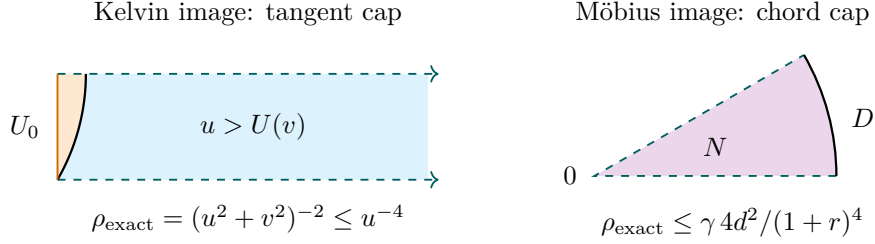


FIGURE 3. The two conformal comparison domains. On the left the thin orange region is added by zero extension across the Dirichlet curve $u = U(v)$ to the Dirichlet line $u = U_0$; the horizontal sides remain Neumann and continue to infinity. On the right the chord and circle become Neumann rays, while the physical Dirichlet side becomes $r = 1$. For visibility the displayed angle is $\pi/6$; the left coordinates are $(u/h_0 - 2 \cot \theta, v/h_0)$, and the right radius is unchanged. Only the written mass inequalities, not the drawn areas, are used in the comparison.

since $33/100 + 1/(4\sqrt{2}) < 51/100$.

Let Δ denote the sum of the model-area excesses of the eight caps. Equations (6.5), (6.22), and (6.26) give

$$\Delta \leq \frac{2}{5}w^2 \tan^3 \theta + \frac{2}{3}\theta g^2 \left(\frac{\gamma}{\beta} - 1 \right). \quad (6.29)$$

If $\gamma/\beta - 1 \leq 3\theta^2/5$, this is at most $\frac{2}{5}(w^2 \tan^3 \theta + \theta^3 g^2)$. The sum of the positive-channel multiplicities, divided by t , is at most

$$2w \tan \theta + \theta g \sqrt{\gamma}. \quad (6.30)$$

If $\mathcal{H}_*$ bounds all the cap logarithmic factors, retaining the constants in (6.27) gives the following total corner excess over the true area term:

$$\begin{aligned} & \sum_{\text{corners}} N(\pi^2 t^2) - \frac{\pi}{4} |\text{all corners}| t^2 \\ & \leq \frac{\pi}{4} \Delta t^2 + t \left[\left(\frac{83}{25} + 2\sqrt{\gamma} \right) g + \frac{116}{25} s \right] + 3 + t(2w \tan \theta + \theta g \sqrt{\gamma}) \mathcal{H}_*. \end{aligned} \quad (6.31)$$

The constant three is four rectangle quarters plus four right-cap halves. For the very thin range it is preferable instead to use (6.28) and $\lfloor v + 1/2 \rfloor \leq 2v$, valid for every $v \geq 0$.

6.10. Closing the all-energy thin range. Assume $0 < \delta \leq 1/50$. Elementary Taylor bounds and feasibility give

$$\theta < \frac{1}{40}, \quad \theta \leq \frac{21}{20}\delta, \quad \tan \theta \leq \frac{53}{50}\delta, \quad w \leq \frac{51}{50}\delta, \quad c, \mathcal{L}(z) \geq \frac{97}{100}, \quad \theta \cot \theta \geq \frac{99}{100}. \quad (6.32)$$

For example $\cos(1/40) + \sin(1/40) > 50/49$, and on this smaller interval $\cos \theta + \sin \theta - 1 \geq (987/1000)\theta$. The remaining estimates follow from $\tan \theta \leq \theta/(1 - \theta^2/2)$, $w \leq \delta + \theta^2/2$, and (6.2). Also

$$\sqrt{\gamma} \leq \frac{1001}{1000}, \quad \frac{\gamma}{\beta} - 1 \leq \frac{3}{5}\theta^2. \quad (6.33)$$

More generally the second bound holds for $z = \theta^2 \leq 1/100$:

$$\gamma \leq (1 - z/4)^{-2} \leq 1 + \frac{51}{100}z, \quad \beta^{-1} \leq (1 - z/36)^{-1/2} \leq 1 + \frac{3}{200}z.$$

Squaring and clearing positive denominators proves these scalar inequalities on that interval, and their product is at most $1 + 3z/5$. The stronger first bound in (6.33) uses $\theta < 1/40$.

Using the absorbed rectangle bound and the bound $2dt'$ for the right zero channel, the linear corner cost is less than $(1017/100)\delta t$. Indeed it is at most $(51/100)4wt + 4wt + 4g\sqrt{\gamma}t$. Furthermore (6.29) and (6.30) give

$$\Delta < \frac{24}{25}\delta^5, \quad 2w \tan \theta + \theta g\sqrt{\gamma} < \frac{161}{50}\delta^2.$$

The first rational comparison is

$$\frac{2}{5} \left(\frac{51}{50}\right)^2 \left(\frac{53}{50}\right)^3 + \frac{2}{5} \left(\frac{21}{20}\right)^3 < \frac{24}{25}.$$

For $t \leq 216\delta^{-3}$ every logarithm argument is below $700\delta^{-2}$. The normalized total corner excess is consequently bounded by

$$F_0(\delta) = \frac{1017}{100}\delta + 163\delta^2 + \frac{161}{50}\delta^2 \left(\frac{1}{2} + \frac{1}{6} \log(700\delta^{-2})\right). \quad (6.34)$$

Each term increases on $(0, 1/50]$. Since $\log(1750000) < 15$ (for instance $e^3 > 20$ gives $e^{15} > 20^5 > 1750000$),

$$F_0(\delta) \leq \frac{2034}{10000} + \frac{163}{2500} + \frac{483}{125000} = 0.272464 < \frac{7}{25}.$$

For the sectors, (6.32) gives $\mathcal{A}(z) \geq 31/100$. Directly from (6.17),

$$\begin{aligned} \mathcal{B}(0) + \mathcal{B}(g) &= \frac{\pi t}{4} \left[\theta \left(c^2 - \frac{g^2}{2} \right) + cg \theta \cot \theta \right] \\ &\geq \frac{47\pi}{200} t(\theta + g) \geq \frac{47\pi}{200} X. \end{aligned}$$

Both bracket coefficients are at least $47/50$, and $\theta + g \geq \delta$ follows from $\cos \theta + \sin \theta - 1 \leq \theta$. The inequality $\min(a, x) + \min(a, y) \geq \min(a, x + y)$ now shows that the normalized sector saving is at least

$$2 \min \left\{ \frac{31}{100}, \frac{47\pi}{200} X \right\} > \frac{7}{20} \quad (X \geq 1/4).$$

This pays the entire corner excess (6.34), so (6.1) holds for $1/(4\delta) \leq t \leq 216\delta^{-3}$.

For the low-energy endpoint, the common-core comparison (4.11) gives

$$N_{Q \setminus H}(\pi^2 t^2) = 0 \quad (X < 1/\sqrt{2}). \quad (6.35)$$

In particular this covers $X < 1/4$. Finally the arithmetic subtraction tail (7.9) applies for $t \geq (5832/125)\delta^{-3}$, hence certainly for $t \geq 216\delta^{-3}$. The ranges overlap and include their joining endpoints. At $E = 0$ the original Dirichlet count is zero. This completes the all-energy assertion for $q \geq 49/50$ at positive angle.

6.11. Closing the remaining high-energy range. We now assume $0 < \delta \leq 2/25$ and $X \geq 8$. Set $K = 5832/125$. By (7.9), only the window

$$\frac{8}{\delta} \leq t \leq K\delta^{-3} \quad (6.36)$$

remains. The geometry (6.3) and the same polynomial bounds used above give

$$\sqrt{\gamma} \leq \frac{1003}{1000}, \quad \frac{\gamma}{\beta} - 1 \leq \frac{3}{5}\theta^2.$$

Every logarithm argument in the cap bounds is at most $\pi \max(w, g\sqrt{\gamma})t < 160\delta^{-2}$. Thus we may put

$$H_\delta = \frac{1}{2} + \frac{1}{6} \log(160\delta^{-2}).$$

The product $\delta^2 H_\delta$ is increasing on $(0, 2/25]$, since its derivative is $\delta[2/3 + \log(160\delta^{-2})/3] > 0$. At the right endpoint, $\log(25000) < 11$, and hence

$$\delta^2 H_\delta < \frac{4}{625} \frac{7}{3}. \quad (6.37)$$

For example $e^3 > 20$ and $e^2 > 7$, from finite positive Taylor sums, give $e^{11} > 56000$. We bound the product in (6.37), not H_δ alone as $\delta \downarrow 0$.

First take $\theta \geq \delta/10$. We already have $\mathcal{A}(z) \geq 2337/10000$. Moreover

$$\mathcal{B}(g) \geq \mathcal{B}(0) = \frac{\pi\theta tc^2}{8} \geq \frac{\pi}{10} \left(\frac{91}{100}\right)^2 > \frac{24843}{100000}.$$

Here the energy hypothesis is exactly $X \geq 8$. The four-sector saving in (6.18), divided by t , is at least $2337/2500$. For the corners, (6.31) gives linear coefficient at most $(2663/500)w \leq (141139/25000)\delta$. The elementary comparisons

$$\frac{2}{5} \left[\left(\frac{53}{50}\right)^2 \left(\frac{6}{5}\right)^3 + \left(\frac{7}{6}\right)^3 \right] < \frac{71}{50}, \quad 2\frac{53}{50}\frac{6}{5} + \frac{7}{6}\frac{1003}{1000} < \frac{93}{25}$$

bound the area and positive-channel coefficients. Since $\pi K/4 < 36.66$, the area contribution divided by t is at most $(521/10)\delta^2$. Also $3/t \leq 3\delta/8$. The total normalized corner error is thus at most

$$\frac{141139}{25000}\delta + \frac{521}{10}\delta^2 + \frac{93}{25}\delta^2 H_\delta + \frac{3\delta}{8}.$$

It increases in δ and, by (6.37), is at most

$$0.4516448 + 0.33344 + 0.055552 + 0.03 = \frac{136037}{156250} = 0.8706368 < \frac{2337}{2500}.$$

The exact residual margin is $20051/312500 > 1/50$.

For $\theta \leq \delta/10$, use instead

$$\tan \theta \leq \frac{1001}{10000}\delta, \quad w \leq \frac{1001}{1000}\delta, \quad \mathcal{L}(z) \geq \frac{919}{1000}, \quad \sqrt{\gamma} \leq \frac{1001}{1000}, \quad g \geq \frac{9}{10}\delta. \quad (6.38)$$

These follow from $\theta \leq 1/125$, $w \leq \delta + \delta^2/200$, and

$$\mathcal{L}(z) \geq q - \frac{\delta^2}{200} - \frac{1001}{20000}\delta^2 > \frac{919}{1000}.$$

The last inequality in (6.38) is $g \geq \delta - \theta$. At the maximal-gap endpoint,

$$\mathcal{A}(g) \geq \frac{7}{20}\frac{919}{1000} - \frac{1001}{1000}\frac{2}{25} = \frac{24157}{100000}.$$

As $b(g) > a(g)$ and $\theta a(g) \geq g$,

$$\mathcal{B}(g) \geq \frac{\pi}{4}\mathcal{L}(g)gt \geq \frac{3}{4}\frac{919}{1000}\frac{9}{10}X \geq 4.9626.$$

Hence $\mathcal{G}(g) \geq 24157/100000$, while $\mathcal{G}(0) \geq 0$. The normalized four-sector saving is at least $24157/50000 = 0.48314$.

The corresponding linear corner coefficient is at most $(2661/500)(1001/1000)\delta = 5.327322\delta$. The area and channel bounds become

$$\Delta \leq \frac{2}{5} \left[\left(\frac{1001}{1000}\right)^2 \left(\frac{1001}{10000}\right)^3 + \frac{1}{1000} \right] \delta^5 < \frac{803}{1000000}\delta^5,$$

and

$$2w \tan \theta + \theta g \sqrt{\gamma} \leq \left[2\frac{1001}{1000}\frac{1001}{10000} + \frac{1}{10}\frac{1001}{1000} \right] \delta^2 < \frac{301}{1000}\delta^2.$$

In (6.36), the area term divided by t is less than $3\delta^2/100$. The normalized corner error is consequently below

$$5.327322\delta + \frac{3}{100}\delta^2 + \frac{301}{1000}\delta^2 H_\delta + \frac{3\delta}{8}.$$

All terms increase in δ . Its endpoint bound is

$$\frac{8641363}{18750000} = 0.460872693333 \dots < \frac{24157}{50000},$$

leaving the exact margin $52189/2343750 > 1/50$.

The two angle ranges overlap at $\theta = \delta/10$ and exhaust all positive feasible rotations. In both ranges, summing the exact piece areas and their bounds proves (6.1); in fact the relaxed mixed count is at most its Weyl term minus $t/50$ throughout (6.36). The arithmetic tail covers every larger t , including the shared endpoint. We have proved the desired high-energy comparison for every $q \geq 23/25$ and $X \geq 8$.

Together with the all-energy result for $q \geq 49/50$, the remaining bounded slab in this size range is

$$\frac{23}{25} \leq q \leq \frac{49}{50}, \quad \frac{1}{4} \leq X \leq 8, \quad t \leq 400.$$

The low-energy zero range follows directly from (6.35). The analytic comparison includes $X = 8$, so it joins the closed finite certificate for this slab without an endpoint gap.

7. AN EXPLICIT CIRCLE ESTIMATE

Write

$$\begin{aligned} G(r) &= \#\{m \in \mathbb{Z}^2 : |m| \leq r\}, \\ C(r) &= \#\{(m, n) \in \mathbb{N}^2 : m^2 + n^2 \leq r^2\}, \\ d(r) &= \frac{\pi r^2}{4} - C(r). \end{aligned}$$

Here and throughout this section $\mathbb{N} = \{1, 2, \dots\}$, and counts include multiplicities and their boundary points. We prove

$$|G(r) - \pi r^2| \leq \frac{71}{10} r^{2/3} \quad (r \geq 64). \quad (7.1)$$

Only the usual integral representation, differential equation, and leading asymptotic terms for J_1 are used; no circle-discrepancy result with an unspecified constant or onset is needed.

7.1. The Bessel bounds and a Euclidean shell sum. The integral representation

$$J_1(x) = \frac{x}{\pi} \int_{-1}^1 e^{ixt} \sqrt{1-t^2} dt$$

and the function and derivative asymptotic expansions in [15, §§10.9, 10.17] give

$$|J_1(x)| \leq x/2, \quad \sqrt{x}J_1(x) = \sqrt{2/\pi} \cos(x - 3\pi/4) + O(x^{-1}),$$

and the corresponding derivative asymptotic is $-\sqrt{2/\pi} \sin(x - 3\pi/4) + O(x^{-1})$. These standard facts imply the uniform bound

$$|J_1(x)| \leq \min\{x/2, (8/9)x^{-1/2}\} \quad (x > 0). \quad (7.2)$$

Indeed the first bound suffices for $0 < x \leq 1$. For $1 \leq x \leq 2$, the alternating power series, whose terms decrease in absolute value, gives a nonnegative lower truncation and

$$0 \leq J_1(x) \leq \frac{x}{2} - \frac{x^3}{16} + \frac{x^5}{384} \leq \frac{x}{2} - \frac{5x^3}{96} \leq \frac{4}{3\sqrt{5}} < \frac{3}{5}.$$

Thus $\sqrt{x}|J_1(x)| < (3/5)\sqrt{2} < 8/9$ on this interval. For $x \geq 2$, put $u(x) = \sqrt{x}J_1(x)$ and $b(x) = 1 - 3/(4x^2)$. The Bessel equation gives $u'' + bu = 0$, so

$$(u'^2 + bu^2)' = b'u^2 \geq 0, \quad \lim_{x \rightarrow \infty} (u'^2 + bu^2) = 2/\pi.$$

In particular

$$|J_1(x)| \leq \sqrt{\frac{2}{\pi}} x^{-1/2} \left(1 - \frac{3}{4x^2}\right)^{-1/2} \quad (x \geq 2). \quad (7.3)$$

Its squared normalized amplitude is at most $32/(13\pi) < 64/81$, which completes (7.2).

Use the Fourier convention $\widehat{f}(\xi) = \int f(x) e^{-2\pi i x \cdot \xi} dx$ and let χ_R be the indicator of the disk of radius R . Radial integration gives

$$\widehat{\chi}_R(\xi) = \frac{RJ_1(2\pi R|\xi|)}{|\xi|}.$$

For $R \geq 63$ and $m \in \mathbb{Z}^2 \setminus \{0\}$, (7.3) consequently gives

$$|\hat{\chi}_R(m)| \leq \frac{100001}{100000} \frac{\sqrt{R}}{\pi} |m|^{-3/2}. \quad (7.4)$$

To check the constant it suffices to use $\pi > 3$, $|m| \geq 1$, and

$$\left(\frac{100001}{100000}\right)^2 \left(1 - \frac{1}{48 \cdot 63^2}\right) > 1.$$

For $K \geq 2$ we next prove

$$F(K) := \sum_{m \in \mathbb{Z}^2 \setminus \{0\}} |m|^{-3/2} \min\{1, (K/|m|)^{3/2}\} \leq 6\pi\sqrt{K}. \quad (7.5)$$

Set $a = 1/\sqrt{2}$, $A(s) = G(s) - 1$, and $P(s) = \pi(s+a)^2 - 1$. The disjoint unit cells centered at the lattice points of a radius- s disk all lie in the radius- $(s+a)$ disk; hence $A(s) \leq P(s)$. There are exactly twelve nonzero lattice points of norm at most two, and their contribution is

$$F_0 = 4 + 4 \cdot 2^{-3/4} + \sqrt{2}.$$

For the continuous decreasing function $f(s) = s^{-3/2} \min\{1, (K/s)^{3/2}\}$, Stieltjes integration by parts, with these twelve points retained exactly, yields

$$\begin{aligned} F(K) &= F_0 - \int_2^\infty (A(s) - 12) f'(s) ds \\ &\leq F_0 + (P(2) - 12) f(2) + \int_2^\infty P'(s) f(s) ds. \end{aligned}$$

All integrals converge, and this equality includes the norm-two points also when $K = 2$. Direct integration gives

$$\int_2^\infty s f(s) ds = 3\sqrt{K} - 2\sqrt{2}, \quad \int_2^\infty f(s) ds = \sqrt{2} - \frac{3}{2\sqrt{K}}.$$

It follows that

$$F(K) \leq 6\pi\sqrt{K} + B - \frac{3\pi}{\sqrt{2K}}, \quad B = 4 + 4 \cdot 2^{-3/4} + \sqrt{2} + 3\pi - \frac{23\pi + 26}{4\sqrt{2}}.$$

The bounds $2^{-3/4} < 119/200$, $\sqrt{2} < 283/200$, and $\pi > 157/50$ give

$$B < 4 + 4 \frac{119}{200} + \frac{283}{200} + 3 \frac{157}{50} - \frac{23(157/50) + 26}{4(283/200)} = -\frac{7831}{56600} < 0.$$

For the direction of these substitutions, the remaining expression is increasing in the variable replacing $\sqrt{2}$, while the coefficient of π is negative even at $283/200$. The two radical inequalities are verified by positive integer powers. This proves (7.5).

7.2. Smoothing and the exact constant. Let $\mu_\varepsilon = \chi_\varepsilon/(\pi\varepsilon^2)$. From (7.2),

$$|\hat{\mu}_\varepsilon(m)| \leq \min\{1, c(\varepsilon|m|)^{-3/2}\}, \quad c = \frac{8/9}{\pi\sqrt{2\pi}}.$$

For $0 < \varepsilon < r$ there is a pointwise convolution sandwich

$$\sum_m (\chi_{r-\varepsilon} * \mu_\varepsilon)(m) \leq G(r) \leq \sum_m (\chi_{r+\varepsilon} * \mu_\varepsilon)(m). \quad (7.6)$$

The lower convolution is zero at an exterior tangency, whereas the upper convolution is one throughout the closed disk of radius r . Thus no boundary-point correction is lost. Both convolutions are continuous and compactly supported. Their periodizations have absolutely summable Fourier coefficients, since the product of the two disk bounds decays as $|m|^{-3}$. Fourier uniqueness therefore identifies the periodization with its absolutely convergent Fourier series at every point, justifying the Poisson equalities in (7.6).

Take $\varepsilon = (3/8)r^{-1/3}$ and $r \geq 64$. Then $r - \varepsilon \geq 63$ and $K = c^{2/3}/\varepsilon > 2$. For the latter assertion, $c > 1/10$, $c^{2/3} > 1/5$, and $\varepsilon \leq 3/32$ give $K > 32/15$. Applying (7.4) and (7.5), the nonzero Fourier terms have total absolute value at most

$$6 \frac{100001}{100000} c^{1/3} \sqrt{\frac{r + \varepsilon}{\varepsilon}}.$$

The zero Fourier terms are the areas of the two smoothing disks. Consequently

$$|G(r) - \pi r^2| \leq 2\pi r\varepsilon + \pi\varepsilon^2 + 6 \frac{100001}{100000} c^{1/3} \sqrt{\frac{r + \varepsilon}{\varepsilon}}.$$

After division by $r^{2/3}$ the right side is

$$\frac{3\pi}{4} + \frac{9\pi}{64}z + 6 \frac{100001}{100000} c^{1/3} \sqrt{8/3 + z}, \quad z = r^{-4/3} \leq \frac{1}{256}.$$

It increases with z . Using

$$\frac{333}{106} < \pi < \frac{355}{113}, \quad c^{1/3} < \frac{4833}{10000}, \quad \sqrt{8/3} < \frac{1633}{1000}, \quad \sqrt{1 + 3z/8} \leq 1 + 3z/16,$$

we bound it above by the exact rational number

$$\begin{aligned} & \frac{3}{4} \frac{355}{113} + \frac{9}{64} \frac{355}{113} + 6 \frac{100001}{100000} \frac{4833}{10000} \frac{1633}{1000} \left(1 + \frac{3}{4096}\right) \\ &= \frac{1642372041329895129}{231424000000000000} < \frac{71}{10}. \end{aligned}$$

The nontrivial radical bound here follows from the rational comparison

$$2(333/106)^3(4833/10000)^6 > (8/9)^2;$$

the others follow by squaring. Also $c > 1/10$ follows from $3200 > 81(355/113)^3$. The two bounds on π follow from the Machin computation described below. This proves (7.1) for every real $r \geq 64$.

7.3. All-real deficit tails. The exact reflection identity $G(r) = 1 + 4[r] + 4C(r)$ gives

$$r - \frac{3}{4} - \frac{71}{40}r^{2/3} \leq d(r) \leq r + \frac{1}{4} + \frac{71}{40}r^{2/3} \quad (r \geq 64). \quad (7.7)$$

In particular

$$(1 - \eta)r \leq d(r) \leq (1 + \eta)r \quad \text{if } r \geq 64, \quad \eta \geq \frac{3}{4r} + \frac{71}{40r^{1/3}}. \quad (7.8)$$

Two rational onsets used below are

η	onset	$\eta - 3/(4v^3) - 71/(40v)$ at the onset v^3
1/19	$34^3 = 39304$	6073/14935520
1/24	$43^3 = 79507$	451/1192605

The right side of the sufficient condition in (7.8) decreases in r .

There is also a convenient uniform tail for $4/5 \leq q < 1$. Writing $\delta = 1 - q$, (7.7) gives, when $t, qt \geq 64$,

$$d(t) - d(qt) \geq \delta t - 1 - \frac{71}{40}(1 + q^{2/3})t^{2/3}.$$

For $t^{1/3} \geq 18/(5\delta)$ its right side is at least $t^{2/3}/20 - 1 > 0$, since $\delta \leq 1/5$. The same hypotheses imply $t, qt > 64$. We have therefore proved

$$d(qt) \leq d(t) \quad \left(\frac{4}{5} \leq q < 1, \quad t \geq \frac{5832}{125(1 - q)^3} \right). \quad (7.9)$$

8. PAIRED-DEFICIT CERTIFICATES ON WHOLE RADIUS INTERVALS

The unsmoothed subtraction test compares $d(qt)$ directly with $d(t)$. Far out, the relative corridors above suffice. On the finite radius range, retaining this paired comparison is stronger than estimating the two deficits by separate relative errors: an outer lower bound can be compared with the largest inner deficit that is actually relevant. The following rational-bin construction implements that observation uniformly over real radii and a whole interval of multipliers q .

Put

$$p_- := \frac{103993}{33102} < \pi < \frac{104348}{33215} =: p_+. \quad (8.1)$$

These enclosures, as well as the weaker bounds used above, are proved by $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$ and the alternating series for $\arctan x$. An even number of terms gives a lower bound; adding the next positive term gives an upper bound. Twelve terms for each arctangent suffice for (8.1). The histogram verifier instead uses twenty-four and eight terms, respectively, and verifies the same strict enclosure. Tangent addition proves the Machin identity with its branch fixed by the elementary interval $0 < \arctan(1/5) < 1/5$.

8.1. Exact bins and prefix comparisons. Let R, h be positive integers. For every unordered positive pair $1 \leq m \leq n$, $m^2 + n^2 \leq R^2$, find the integer j satisfying

$$(j-1)^2 < h^2(m^2 + n^2) \leq j^2. \quad (8.2)$$

Add weight two in bin j when $m < n$, and weight one when $m = n$. The prefix sums of these bins are exactly $H_j = C(j/h)$, $0 \leq j \leq hR$. In particular lattice points on a bin boundary are counted inclusively. A floating square root is used only to propose j ; both integer inequalities (8.2) are checked for every pair, and verification stops if either fails. Thus floating tolerances make no proof decision. The diagonal correction ensures that all ordered-pair multiplicities are retained.

For the closed interval $j/h \leq t \leq (j+1)/h$, monotonicity gives

$$\frac{p_- j^2}{4h^2} - H_{j+1} \leq d(t) \leq \frac{p_+ (j+1)^2}{4h^2} - H_j.$$

Use the following integer lower and upper bounds in units of $1/(4h^2)$:

$$L_j = \left\lfloor p_- j^2 - 4h^2 H_{j+1} \right\rfloor, \quad U_j = \left\lceil p_+ (j+1)^2 - 4h^2 H_j \right\rceil. \quad (8.3)$$

Form the prefix maxima $M_j = \max_{0 \leq i \leq j} U_i$. For $0 < q \leq q_0 < 1$, the number qt belongs to an inner bin with index at most $\lfloor q_0(j+1) \rfloor$. It follows that the single integer comparison

$$L_j \geq M_{\lfloor q_0(j+1) \rfloor} \quad (8.4)$$

proves $d(qt) \leq d(t)$ for every t in this closed outer bin and every $0 < q \leq q_0$. The prefix includes the last index even when $q_0(j+1)$ is an integer. This deliberate inclusion resolves both the radius and the multiplier endpoints.

The verifier tests all bins $h \leq j < hR$. If j_* is the last failed bin, it reports $(j_* + 1)/h$ as the start of the certified suffix. Every bin of that suffix is checked, not merely its endpoints. If the last bin fails it reports no suffix; it does not claim an unchecked singleton at R . The integer-prefix proof is therefore a proof over real intervals, not a collection of successful radius samples.

8.2. Executed ranges and their analytic continuation. The three executed histograms have the following parameters. The pair column counts unordered positive pairs, before their multiplicities are assigned.

R	h	pairs	histogram bytes
50000	100	981740308	40000008
5000	1000	9816727	40000008
90000	100	3180849242	72000008

The first run proves the finite portions of the following infinite tails; in this table $4/5 \leq q \leq q_0$ and $t \geq T$:

q_0	T	
81/100	13733/100	
41/50	725/4	
83/100 or 21/25	977/4	$d(qt) \leq d(t).$
17/20 or 43/50	38427/100	
7/8	2317/4	
9/10	67111/50	

(8.5)

Indeed (8.4) proves each statement up to 50000. For $t \geq 50000$ and $4/5 \leq q \leq 9/10$, both t and qt are at least $40000 > 39304$. The $\eta = 1/19$ corridor gives

$$d(qt) \leq \frac{20}{19}qt \leq \frac{18}{19}t \leq d(t).$$

This joins the finite and infinite intervals inclusively at 50000.

For $q_0 = 23/25$ the finer run proves the interval $[163542/125, 5000]$, and the radius-90000 run proves $[148571/50, 90000]$. They overlap since $148571/50 < 5000$. For $t \geq 90000$ and $9/10 \leq q \leq 23/25$, both radii are at least $81000 > 79507$, so the $\eta = 1/24$ corridor gives $d(qt) \leq (25/24)qt \leq (23/24)t \leq d(t)$. Thus

$$d(qt) \leq d(t) \quad \left(\frac{9}{10} \leq q \leq \frac{23}{25}, \quad t \geq \frac{163542}{125} \right). \quad (8.6)$$

The lower restriction on q is retained in this analytic continuation; the wider finite-prefix statement is not silently extended to infinity.

8.3. Arithmetic safety and independent checks. The histogram and prefix arrays use signed 64-bit integers. The largest base product in the three runs is

$$104348(90000 \cdot 100)^2 = 845218800000000000 < 2^{63} - 1 = 9223372036854775807.$$

The radius-5000 run has $hR = 5000000$, giving the smaller bound 2608700000000000000 . Count-dependent products fit the same bound before subtraction: the cells $[m-1, m] \times [n-1, n]$ belonging to positive lattice points inside a radius- r disk lie inside its positive quadrant, whence $C(r) \leq \pi r^2/4$. For example $4h^2 \cdot 33215 C(r) < 104348(hr)^2$. Additional products for the relative-corridor options, the rational q_0 index, and ceiling operations have explicit range checks before use. Counts in the largest run exceed 32-bit range; they are never stored in 32-bit count arrays.

In each full run, thirty-two dispersed rational radii, together with the endpoints specified by the code, are checked against the independent integer-row identity

$$C(j/h) = \sum_{m=1}^{\lfloor \sqrt{B} \rfloor} \lfloor \sqrt{B - m^2} \rfloor, \quad B = \lfloor j^2/h^2 \rfloor.$$

These checks supplement, but do not replace, the pairwise bin validation. An independent small-radius audit reconstructs every ordered pair and every rounded bin comparison for all sixteen choices $R \in \{2, 3, 17, 61\}$, $h \in \{1, 3, 7, 100\}$, totaling 9229 bins. It also tests the failed-last-bin convention. For each of the three parameter choices, the large histogram was generated and checked in a full run. The small audit is not described as an independent repetition of billions of pairs.

9. FINITE INDEXED CERTIFICATES FOR THE CONTINUUM OF SQUARE HOLES

Normalize the outer square to side one. A hole of side q has area q^2 , regardless of its feasible angle and strictly interior position. All eigenvalues in this section are those of the ordinary Euclidean Dirichlet Laplacian. The comparison maps used below supply Rayleigh quotient inequalities for this operator; no pulled-back metric is substituted into the theorem.

9.1. From the count deficit to a finite indexed obligation. We use the reference estimate (4.3) and its exact integer form (4.24). In the radius notation of this section, $C_n = C(\sqrt{n})$ and $j_r(\tau) = C_{\tau-1} - C_{\lfloor r^2\tau \rfloor} + 1$. The outer count is strict and the inner count inclusive, including at multiple eigenvalues. The concentric-enlargement argument of Section 4.1 makes this a uniform indexed estimate for every reference angle and strictly interior position. Dirichlet insertion also gives

$$N_{Q \setminus H}(\pi^2 t^2) + C(qt) \leq C(t). \quad (9.1)$$

Thus $d(qt) \leq d(t)$ implies the inclusive Weyl bound $N_{Q \setminus H}(\pi^2 t^2) \leq \pi(1 - q^2)t^2/4$.

Suppose an inclusive count bound is proved for all energies at least $E_0(q)$. At the onset it bounds the number of eigenvalues at or below $E_0(q)$ by $\lfloor (1 - q^2)E_0(q)/(4\pi) \rfloor$. Every larger index has its eigenvalue strictly above the onset, where the count bound applies at that eigenvalue itself. It is therefore enough to verify the remaining finite indices. This reasoning includes multiple eigenvalues.

For a closed size interval $[l, h]$, the three cutoffs used here are

proved radius onset $T(q), \quad E_0(q) = \pi^2 T(q)^2$	sufficient largest finite index
$201/q$	$\lfloor p_+(1 - l^2)201^2/(4l^2) \rfloor$
T independent of q	$\lfloor p_+(1 - l^2)T^2/4 \rfloor$
$8/(1 - q)$	$\lfloor 16p_+(1 + h)/(1 - h) \rfloor$

(9.2)

The first uses the scalar corridor $8u/9 \leq d(u) \leq 10u/9$ for $u \geq 201$, and is used only for $q \leq 4/5$. The second uses (8.5) or (8.6). The third uses the proved geometric count estimate for $q \geq 23/25$, $(1 - q)t \geq 8$. Its Weyl-index threshold $16\pi(1 + q)/(1 - q)$ increases with q ; the right endpoint h , not l , is essential in this row.

9.2. Interval comparisons and exact finite tests. For most records the reference is the same-center, same-angle hole of side $r = l$. Since $r \leq q$, actual set inclusion gives the comparison factor one. Put $S = 2^{40}$ and

$$\mu(l, h) = \left\lfloor \frac{Sp_-}{4}(1 - h^2) \right\rfloor. \quad (9.3)$$

For every target k , it suffices to find an integer $1 \leq \tau \leq M$ with

$$j_l(\tau) \leq k, \quad \mu(l, h)\tau \geq Sk. \quad (9.4)$$

Indeed $(1 - q^2)\lambda_k/(4\pi) \geq p_-(1 - h^2)\tau/4 \geq k$ throughout the whole closed interval. The reference count is uniform over position and angle, so these variables are not sampled.

The generators form a cumulative maximum of admissible energies for each gap index and all larger target indices. The independent replays for $7/10 \leq q \leq 49/50$ use a different construction. They form the suffix minimum

$$B_r(n) = \min_{n \leq \tau \leq M} j_r(\tau), \quad n_k = \left\lceil \frac{Sk}{\mu(l, h)} \right\rceil,$$

check $1 \leq n_k \leq M$, and test $B_l(n_k) \leq k$ for every k . This is exactly the existence assertion (9.4). The finite bound M is only the search limit for reference energies; the analytic onset, not M , supplies the infinite spectral tail.

The first two indices can be supplied directly. Domain monotonicity gives $\lambda_1(\Omega) \geq 2\pi^2$ and $\lambda_2(\Omega) \geq 5\pi^2$. Thus $q \leq 1/2$ implies $(1 - q^2)\lambda_1 \geq 3\pi^2/2 > 4\pi$, and $q \leq 7/10$ implies $(1 - q^2)\lambda_2 \geq 51\pi^2/20 > 8\pi$; the last comparison follows from $\pi > 333/106 > 160/51$. For $q > 1/2$ the common-core ratio in (4.11) exceeds $3\pi/8 > 1$, and for $q > 7/10$ it exceeds $17\pi/24 > 2$. This completes both indices without further geometric input. The classical Faber–Krahn and Krahn–Szegő inequalities give an alternative general route to these two indices; see [13].

For $q > 1/2$, the all-position common-core estimate (4.11) also permits every index

$$k \leq \max \left\{ 2, \left\lfloor \frac{p_-(1 + l)}{8(1 - l)} \right\rfloor \right\}. \quad (9.5)$$

The lower-band larger-reference coefficient is (4.25), with $K_* = 99/70$ and the strict feasibility check $K_* \max\{h, r\} < 1$. The fifth-mode flag is precisely (4.28); the three angular repairs are those of Section 4.6. These are independent alternatives to (9.4), and their applicability is recomputed for each closed size interval rather than trusted from a saved flag.

9.3. The completed finite census and the joins. All size endpoints are rational with denominators dividing 64000. The following table records completed finite checks, before adding their analytic tails. The “pairs” column is the number of interval/index obligations, including those discharged by a stated analytic low-mode bypass; it is not a count of computed eigenvalues.

closed size range	intervals	pairs	largest k	largest M
[12/25, 7/10]	315	17944637	105989	250000
[7/10, 4/5]	267	6343627	33026	250000
[4/5, 83/100]	120	1888608	16867	250000
[83/100, 9/10]	1241	261974004	331626	2000000
[9/10, 49/50]	1692	241379723	255435	2000000

(9.6)

The second row has fifteen residual interval/index pairs, exactly the $k = 9$ and $k = 11$ bands in Section 4.6. The third has ten, exactly its $k = 5$ band. The rotation repairs close all twenty-five. The last two rows have no residual finite pair. The first row's twenty-eight extra references and its permitted Gram bypass are already included in its successful replay.

For clarity, the tail and local search cap in each upper subband are listed explicitly. No saved interval crosses one of these changes.

$[l_0, h_0]$	$T(q)$	M	intervals	pairs
[4/5, 83/100]	977/4	250000	120	1888608
[83/100, 21/25]	977/4	250000	40	567818
[21/25, 43/50]	38427/100	250000	160	5150545
[43/50, 7/8]	2317/4	500000	240	15651951
[7/8, 9/10]	67111/50	2000000	801	240603690
[9/10, 23/25]	163542/125	2000000	1051	239521973
[23/25, 49/50]	$8/(1 - q)$	250000	641	1857750

(9.7)

In particular the first row uses 977/4 on its entire interval, not the shorter optional tails for $q \leq 81/100$ or $q \leq 41/50$. At $q = 9/10$ both paired-tail arguments apply. At $q = 23/25$ the constant-tail row and the $X = (1 - q)t \geq 8$ row both include the endpoint; the latter uses the third cutoff in (9.2). At $q = 49/50$ the finite band meets the all-energy thin-hole argument.

The remaining lower interval is covered by the orientation-independent small-hole result: its threshold $\sqrt{1 - 12/(5\pi)}$ is strictly larger than 12/25. For $q \leq 4/5$ the radius-201 corridor supplies the tail in the first row of (9.2); for $4/5 \leq q \leq 23/25$ the exact paired tails above supply it; for $23/25 \leq q \leq 49/50$ the geometric $X \geq 8$ bound supplies it. The all-energy argument for $49/50 \leq q < 1$ completes the size interval. The zero-angle endpoint is included by the rotation-continuity argument in Section 4.8. Thus the finite computation does not require a separate assumption about parallel square holes or a positive lower bound on boundary clearance.

9.4. Reconstruction and reproducibility – (Github Repository.) Every independent finite replay constructs the complete sorted multiset

$$\{m^2 + n^2 : m, n \geq 1, m^2 + n^2 \leq M\}$$

and obtains C_n by inclusive ordered search. There are 195837 entries at $M = 250000$ and 1569392 at $M = 2000000$, with repetitions retained. Direct integer-root row sums check selected radii as a second construction. Each replay recomputes its cutoff, downward coefficient, reference count, low-index bypass, and residual list. It requires the first and last size endpoints to be the prescribed endpoints and each successive left endpoint to equal the preceding right

endpoint. It then checks every target index, not a sample, and checks the aggregate counts in (9.6).

The bounded array computations use signed 64-bit integers. In these replays $M \leq 2000000$, $S = 2^{40}$, and denominators are at most 64000; the bounds $SM < 2^{63}$ and $64000^2 M < 2^{63}$ are checked before use. Also $0 < \mu < S$, and all actual index/energy products obey the same bound. The ceiling computation uses $Sk + \mu - 1$, which is still below 2^{63} for the displayed maximum k . Scalar fractions and angular root enclosures use arbitrary-precision integers. All acceptance guards raise explicit exceptions rather than using assertions disabled by Python’s optimization switch.

The verification supplement consists of executable source and rational JSON records. The entry points for the five rows of (9.6), in order, are

```
replay_all_angles_band.py
replay_midband_independent.py
replay_upper_midband_independent.py
replay_compact_band_independent.py
replay_final_compact_independent.py
```

Their corresponding data files are

```
all_angles_certificate.json
certificate_midband.json
certificate_upper_midband.json
certificate_compact.json
certificate_final.json.
```

These files contain exact witness inputs. The independent replay reconstructs every residual list and verifies that the specified repairs close it. The last replay imports only the elementary checking helpers and rational π bounds of the compact replay; none imports a certificate generator or an exploratory search.

The scalar and angular entry points are, respectively,

```
verify_circle_tail_constants.py
verify_rotation_gram.py
verify_low_mode_rotation_repairs.py
verify_fifth_mode_upper_repair.py.
```

They use only Python’s standard library and require no certificate data as input. Both angular verifiers reconstruct their closed covers and write the resulting rational leaf records. The histogram source is `certify_square_corridor_histogram.py`, with its independent small audit `audit_square_corridor_histogram_small.py`. The finite lattice replays and histogram programs additionally require NumPy; no numerical eigensolver, Bessel library, or arbitrary-precision floating package is used.

All listed finite and angular checks have been executed successfully with `python -B -O`. The circle constant verifier checks the displayed rational constants; the analytic Bessel, Fourier, and operator arguments remain part of the written proof, not something purportedly certified by that program. Similarly, the finite replays check the stated gates and do not prove the geometric $X \geq 8$ estimate by computation.

The supplement README specifies the three histogram command lines and the remaining execution order. The commands use the three (R, h) pairs displayed above, the paired-prefix option, and exact rational corridor parameters. The reports store the verified parameters and endpoints, not the full histogram; reproduction regenerates all bins. The largest run processes over three billion unordered pairs in vectorized rows and allocates arrays of order hR ; the histogram alone uses 72000008 bytes, with additional arrays for rounded bounds and prefixes. These resource requirements are explicit and finite. The small-hole and radius-201 scalar certificates have their separate specifications and entry points; they are not replaced by any unexecuted extrapolation of the new histograms.

10. BOUNDARY CONTACT BY ONE-SIDED APPROXIMATION

The interior-hole arguments now give the full statements announced in the introduction, including contact. This extension is an immediate consequence of domain monotonicity; it requires neither a new local boundary estimate nor a spectral-continuity assertion at a degenerating configuration.

Let H be a closed square of positive side length contained in $\overline{Q}$, and suppose $\Omega = Q \setminus H$ has positive area. The center z of H belongs to Q . For $0 < \varepsilon < 1$ put

$$H_\varepsilon = z + (1 - \varepsilon)(H - z), \quad \Omega_\varepsilon = Q \setminus H_\varepsilon.$$

Each H_ε is strictly contained in Q , while $\Omega \subset \Omega_\varepsilon$. The Dirichlet form on $H_0^1(\Omega)$ has compact resolvent even when Ω is disconnected: extension by zero into a containing square gives the compact embedding into L^2 . Extension by zero also proves the usual domain monotonicity for these bounded open sets. Hence the interior conclusion gives, for every $k \geq 1$,

$$|\Omega_\varepsilon| \lambda_k^D(\Omega) \geq |\Omega_\varepsilon| \lambda_k^D(\Omega_\varepsilon) \geq 4\pi k. \quad (10.1)$$

As $\varepsilon \downarrow 0$, the areas converge to $|\Omega|$. This proves (1.2) for every allowed contact configuration, without taking a limit of eigenvalues.

For finitely many closed square holes contained in $\overline{Q}$ with pairwise disjoint interiors, shrink each about its own center by the common factor $1 - \varepsilon$. The resulting closed holes lie in their respective disjoint interiors and strictly inside Q ; being a finite family of disjoint compact sets, they have positive separations. Their total side length still satisfies the hypothesis of (1.6). The shrunk-hole complement contains the original complement, and its area converges to the original area. The same inequality (10.1) therefore proves the multiple-hole assertion with outer-boundary and mutual contact. Intersections of the original boundaries have area zero, so the complementary area is still the outer area minus the sum of the individual square areas. The total-side hypothesis ensures that this area is positive. No connectedness assumption is needed.

11. LARGE PARALLEL RECTANGULAR HOLES WITH ARBITRARY DISPLACEMENT

The last geometric conclusion allows the two sides of the hole to vary independently. Let the outer square have side one and let the strictly interior parallel rectangle have side lengths $p, q \in [11/12, 1)$. Its position is arbitrary. We prove

$$N_\Omega(E) \leq \frac{1 - pq}{4\pi} E \quad (E \geq 0).$$

Scaling and evaluation at each eigenvalue then give (1.9). This argument does not use the disk-discrepancy estimate.

Rotate the whole configuration by a quarter turn, if necessary, so that $p \leq q$, with p horizontal. Put

$$\delta = 1 - q, \quad c = \frac{1 - p}{1 - q} \geq 1, \quad x = \delta \frac{\sqrt{E}}{\pi}, \quad z = cx.$$

The two horizontal gaps have lengths δa and $\delta(1 - a)$, and the side gaps have lengths $c\delta b$ and $c\delta(1 - b)$, for independent $a, b \in (0, 1)$. The width-indexed sums A_h, R_h and $R = R_1$ are those defined in (5.1).

Cut the frame into its two full-width horizontal rectangles and two middle side rectangles. Relax matching at the cuts and the entire inward horizontal edge of each horizontal rectangle to Neumann. Its other three edges remain Dirichlet. The side rectangles are Dirichlet across their widths and Neumann at both ends of their length q . This enlarges the original Dirichlet form domain, with norm and energy unchanged on restrictions.

The horizontal transverse frequencies are $(j + 1/2)/(\delta a)$ and $(j + 1/2)/(\delta(1 - a))$; summing their longitudinal Dirichlet counts and bounding floors gives $\delta^{-1}(A_a + A_{1-a})$. Each side

transverse frequency contributes one plus a floor, with the one counting the longitudinal Neumann zero mode even at equality. For every $s \geq 0$,

$$\lfloor bs \rfloor + \lfloor (1-b)s \rfloor \leq \lfloor s \rfloor.$$

Integration against $s/\sqrt{x^2 - s^2}$ gives $R_b(x) + R_{1-b}(x) \leq R(x)$, because a frequency $\nu \leq x$ contributes $\sqrt{x^2 - \nu^2}$. Applying this comparison to total side width c , and retaining the zero modes, yields

$$\delta N_\Omega(E) \leq A_a(x) + A_{1-a}(x) + \frac{q}{c} R(cx) + \delta \lfloor cx \rfloor. \quad (11.1)$$

Here $R_c(x) = R(cx)/c$. Since $1 - pq = \delta(1 + qc)$, the desired Weyl expression, multiplied by δ , is $\pi(1 + qc)x^2/4$.

An extremal reduction of the side lengths. For fixed c, x, a , let

$$D = A_a(x) + A_{1-a}(x) + \frac{1-\delta}{c} R(cx) + \delta \lfloor cx \rfloor - \frac{\pi}{4} (1 + (1-\delta)c)x^2. \quad (11.2)$$

The decreasing right-endpoint sum satisfies $R(u) \leq \pi u^2/4$. Consequently

$$\frac{\partial D}{\partial \delta} = \lfloor cx \rfloor + \frac{1}{c} \left(\frac{\pi}{4} (cx)^2 - R(cx) \right) \geq 0.$$

The condition $p \geq 11/12$ gives $\delta \leq 1/(12c)$. It therefore suffices to take

$$\delta = \frac{1}{12c}, \quad p = \frac{11}{12}, \quad q = 1 - \frac{1}{12c}. \quad (11.3)$$

This is a scalar comparison at fixed x ; no geometric inclusion is asserted.

For $x > 0$ and $z = cx \geq x$, put

$$\begin{aligned} d_R(z) &= \frac{\pi}{4} z^2 - R(z), \\ \mathcal{H}(x, a) &= \frac{A_a(x) + A_{1-a}(x)}{x} - \frac{\pi}{4} x, \\ \mathcal{K}(z, x) &= \left(1 - \frac{x}{12z} \right) \frac{d_R(z)}{z} - \frac{\lfloor z \rfloor}{12z}, \\ \mathcal{L}(z) &= \frac{11}{12} \frac{d_R(z)}{z} - \frac{\lfloor z \rfloor}{12z}. \end{aligned} \quad (11.4)$$

In (11.3), $D/x = \mathcal{H} - \mathcal{K}$. Moreover $d_R \geq 0$ and $x \leq z$ give

$$\mathcal{K}(z, x) \geq \mathcal{L}(z). \quad (11.5)$$

Scalar estimates with explicit tails. Section 5 proved the uniform midpoint estimate (5.4), with its finite certificate in Appendix B. The scaling identity $A_h(x) = A_1(hx)/h$ now gives

$$\mathcal{H}(x, a) \leq \frac{8}{25} \quad (x > 0, 0 \leq a \leq 1). \quad (11.6)$$

The integer-point version of the same trapezoid argument is (5.2); its initial endpoint contributes $-y/2$. Using that estimate and $\lfloor z \rfloor \leq z$ gives

$$\mathcal{L}(z) \geq \frac{3}{8} - \frac{11}{12} \sqrt{\frac{2}{27}} z^{-1/2} \quad (z \geq 1). \quad (11.7)$$

The finite certificates in Appendix B and this tail prove

$$\mathcal{L}(z) \geq \frac{7}{25} \quad (z \geq 1), \quad \mathcal{L}(z) \geq \frac{8}{25} \quad (z \geq 4). \quad (11.8)$$

Their finite ranges are $[1, 7)$ and $[4, 21)$, respectively. For the two tails, the exact positive differences are

$$7 \left(\frac{3}{8} - \frac{7}{25} \right)^2 - \left(\frac{11}{12} \right)^2 \frac{2}{27} = \frac{9061}{9720000},$$

and

$$21 \left(\frac{3}{8} - \frac{8}{25} \right)^2 - \left(\frac{11}{12} \right)^2 \frac{2}{27} = \frac{12463}{9720000}.$$

A further compact certificate gives

$$\mathcal{H}(x, a) \leq \frac{4}{15} \quad (2 \leq x \leq 4, 0 \leq a \leq 1). \quad (11.9)$$

Concavity removes displacement at low energies. For $1 \leq x \leq 2$ there is a more precise structural bound:

$$A_a(x) + A_{1-a}(x) \leq \max\{A_1(x), 2\sqrt{x^2 - 1}\}. \quad (11.10)$$

If only one width contributes, monotonicity in width bounds its sum by $A_1(x)$. If both contribute nonzero terms, a second frequency in either width would require total scaled width strictly greater than $1/2 + 3/2 = 2$. Hence each width contributes only its first nonzero term. The function $h \mapsto \sqrt{x^2 - 1/(4h^2)}$ is concave on its active interval. The symmetric sum is therefore maximized at $a = 1/2$, giving the second term in (11.10). Additional terms at $x = 2$ have value zero, so the endpoint is included.

Write

$$h_0(x) = 2\sqrt{1 - x^{-2}} - \frac{\pi}{4}x \quad (1 \leq x \leq 2).$$

Then (5.4) and (11.10) imply $\mathcal{H} \leq \max\{4/25, h_0\}$. The last finite certificate is the explicit comparison

$$h_0(x) \leq \mathcal{K}(z, x) \quad (1 \leq x \leq 2, x \leq z < 4). \quad (11.11)$$

Only the two spectral scales remain; no displacement variable occurs in this certificate.

Completion of the geometric proof. If $z \geq 4$, equations (11.6), (11.8) and (11.5) give $\mathcal{H} \leq \mathcal{K}$. Suppose $z < 4$, so $x \leq z < 4$. For $2 \leq x < 4$, use $\mathcal{H} \leq 4/15 < 7/25 \leq \mathcal{L} \leq \mathcal{K}$. For $1 \leq x \leq 2$, (11.11) treats the h_0 branch, while $4/25 < 7/25 \leq \mathcal{K}$ treats the other branch.

If $0 < x < 1$, both horizontal widths cannot contribute nonzero terms, so $A_a + A_{1-a} \leq A_1(x)$ and $\mathcal{H} \leq 4/25$. When $z \geq 1$, the lower bound $\mathcal{K} \geq 7/25$ finishes the comparison. If $z < 1$, both $R(z)$ and $\lfloor z \rfloor$ vanish, giving

$$\mathcal{K}(z, x) = \frac{\pi}{4}z - \frac{\pi}{48}x \geq \frac{11\pi}{48}x.$$

For $x < 1/2$ one has $\mathcal{H} = -\pi x/4 < 0$; for $1/2 \leq x < 1$ one has $\mathcal{K} \geq 11\pi/96 > 4/25$. The case $x = 0$ is the zero-energy count and requires no division. The stated closed ranges cover $x = 1/2, 1, 2, 4$ and $z = 1, 4$; the other integer z boundaries are covered in the finite certificate by the interval beginning there.

Thus $D \leq 0$ at every energy and for every allowed position and pair of side lengths. Equation (11.1) proves the required inclusive count inequality and hence (1.9). This rectangular-hole conclusion does not cover arbitrary intermediate pairs of side lengths, rotated nonsquare rectangular holes, or an arbitrary outer rectangle.

APPENDIX A. EXACT ARITHMETIC AND THE DISK-DEFICIT CERTIFICATES

This appendix specifies the finite arithmetic inputs to Section 3 and the corridor (3.16). The analytic reductions in those sections and Section 7 are part of the proof; no finite enumeration alone asserts an all-energy conclusion.

Rational constants and inclusive lattice counts. The Machin identity

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$$

and the alternating series

$$\arctan(1/a) = \sum_{j \geq 0} \frac{(-1)^j}{(2j+1)a^{2j+1}}$$

give rational enclosures. An even partial sum is a lower bound, and adding the next positive term gives an upper bound. Twelve terms for each arctangent suffice for both

$$\frac{333}{106} < \pi < \frac{355}{113}, \quad \pi_- := \frac{103993}{33102} < \pi < \frac{104348}{33215} =: \pi_+. \quad (\text{A.1})$$

For completeness, tangent addition gives $\tan(4 \arctan(1/5) - \arctan(1/239)) = 1$ exactly. The same series bounds place that angle between zero and one, hence in $(0, \pi/2)$, fixing the branch $\pi/4$.

For an integer cutoff M , form

$$r_n = \#\{(a, b) \in \mathbb{N}^2 : a^2 + b^2 = n\}, \quad C_n = \sum_{j=0}^n r_j \quad (0 \leq n \leq M). \quad (\text{A.2})$$

Enumerating unordered pairs and assigning weight two off the diagonal and one on it retains all multiplicities. Alternatively, enumerate every ordered pair directly. The identity

$$C_n = \sum_{a=1}^{\lfloor \sqrt{n} \rfloor} \left\lfloor \sqrt{n - a^2} \right\rfloor \quad (\text{A.3})$$

provides direct row checks. All root floors here are exact integer square roots. The small-hole verifier uses $M = 1000000$, for which $C_M = 784387$.

Upper bounds use left limits; lower bounds use inclusive values. Between consecutive integer squared radii, $d(u)/u = \pi u/4 - C_n/u$ increases. To bound it above, the finite tests therefore use left limits at the next possible jump. Put $U_n = 355n - 452C_{n-1}$. Whenever $U_n > 0$, check

$$9U_n^2 \leq 1808^2 n \quad (1 \leq n \leq 1000000), \quad (\text{A.4})$$

$$(1000U_n)^2 \leq (1309 \cdot 452)^2 n \quad (1 \leq n \leq 18). \quad (\text{A.5})$$

If $U_n \leq 0$ the upper comparison is immediate, without squaring. These tests establish the finite parts of (3.4); (3.3) covers every $u \geq 1000$, including the last endpoint.

For lower bounds put $L_n = 333n - 424C_n$, now using the inclusive count. In addition to (3.8) and (3.9), check

$$L_n \geq 0, \quad (500000L_n)^2 \geq (318087 \cdot 424)^2 n \quad (21 \leq n \leq 76, n \neq 53), \quad (\text{A.6})$$

$$L_n \geq 0, \quad (125L_n)^2 \geq (81 \cdot 424)^2 n \quad (76 \leq n \leq 1000000). \quad (\text{A.7})$$

The sole failure of the middle test with $n = 53$ included is exactly that index. The partition argument (3.12)–(3.15) supplies its whole energy band. It is not omitted on numerical grounds. Monotonicity between jumps, the overlapping transition intervals in Section 3, and (3.3) give the full real ranges.

The program `verify_deficits.py` reconstructs these counts and tests, the rational bounds placing s_* and its three transitions, the five partition endpoint bounds, and the weakened tail constants used in (3.3). It takes no certificate data as input. Its arithmetic is standard-library integer and rational arithmetic.

The corridor beginning at radius 201. The program `verify_corridor.py` constructs the complete positive-quadrant count through $M = 15625^2 = 244140625$. Frequencies are accumulated in segments of at most 4000000 unsigned 32-bit entries; cumulative counts carry between adjacent segments. Thus the frequency segment occupies at most 16 MB, not an array proportional to the full cutoff.

On $n \leq u^2 < n + 1$, the lower and upper endpoint tests for (3.16) are the conservative integer inequalities

$$9(103993n - 4 \cdot 33102C_n) \geq 32 \cdot 33102 \lceil \sqrt{n} \rceil, \quad (\text{A.8})$$

$$9(104348(n + 1) - 4 \cdot 33215C_n) \leq 40 \cdot 33215 \lfloor \sqrt{n + 1} \rfloor. \quad (\text{A.9})$$

The first tests the inclusive lower endpoint, rounding the required right side upward. The second tests the left limit at the upper endpoint, rounding its permissible right side downward. The use of rational lower and upper bounds for π has the same conservative directions.

The last lower failure occurs at $n = 17226$, and the last upper failure at $n = 40320$. Both are below $201^2 = 40401$. The final count is $C_{244140625} = 191731890$. All subsequent finite bands pass. The analytic tail in the proof of (3.16) begins at radius 15625, so there is no unverified endpoint between the finite and infinite parts. The verifier explicitly requires the certified onset to be at most 201.

For speed, the corridor verifier uses NumPy integer arrays. A floating square root is used only to propose an integer r ; the inequalities $r^2 \leq n < (r+1)^2$ are then checked exactly before the integer root is used. Failure of that validation stops verification. Floating-point tolerances do not decide any proof inequality. Array sizes, cumulative counts and all fixed-width integer products are bounded before use; scalar rational calculations use arbitrary-precision Python integers.

APPENDIX B. EXACT CERTIFICATES FOR THE RECTANGULAR-HOLE COMPARISON

The accompanying program `verify_rectangular.py` verifies the five finite statements in Section 11. It is independent of the other verifiers and uses only integer and rational arithmetic. Twelve-term alternating arctangent enclosures in the Machin identity recheck $333/106 < \pi < 355/113$.

Put $P = 333/106$ and $B_0 = 2^{24}$. For a nonnegative rational number $r = u/v$, define

$$U(r) = \text{isqrt} \left(\left\lfloor \frac{uB_0^2}{v} \right\rfloor \right) + 1, \quad T(r) = U(r) - 1. \quad (\text{B.1})$$

Then $T(r)/B_0 \leq \sqrt{r} < U(r)/B_0$, also when $r = 0$ or the radical is a perfect square. In the following acceptance tests, every upper sum uses these enclosures after monotonicity has moved its energy and width arguments to the indicated endpoints. All failed comparisons cause subdivision; exhaustion of a stated grid or depth is a fatal verification failure.

The midpoint bound and the lower envelopes. For (5.4), use squared energy $r = y^2$ and begin with $[1/4, 9]$. A rational interval $[l, u]$ is accepted if

$$\frac{P}{4}l + \frac{4}{25} \frac{T(l)}{B_0} \geq \text{upper } A_1(\sqrt{u}). \quad (\text{B.2})$$

Otherwise bisect it. The 123 accepted intervals have total width $35/4$, giving the full closed finite range $1/2 \leq y \leq 3$.

For a lower bound $\mathcal{L} \geq \gamma$, divide the z axis into half-open integer bands $n \leq z < n+1$. On a subinterval $[l, u]$ accept if

$$\frac{11P}{4}l^2 \geq 11 \text{ upper } R(u) + n + 12\gamma u. \quad (\text{B.3})$$

This is obtained by multiplying the desired inequality by $12z$. Take $\gamma = 7/25$ and $n = 1, \dots, 6$, or $\gamma = 8/25$ and $n = 4, \dots, 20$. Rational bisection produces accepted widths summing to six and seventeen, respectively. Their final endpoints are covered by (11.7). At an integer boundary, the next band uses the new floor value.

The remaining horizontal range. For (11.9), let $r = x^2$ and use grid denominators $G_0 = K_0 = 2^{14}$ for r and a . The initial rectangle is $[4, 16] \times [0, 1/2]$; symmetry gives the remaining displacement range. A box $[l, u] \times [a, b]$ is accepted if

$$\frac{P}{4}l + \frac{4}{15} \frac{T(l)}{B_0} \geq \text{upper}\{A_b(\sqrt{u}) + A_{1-a}(\sqrt{u})\}. \quad (\text{B.4})$$

Both width sums increase in their width and squared energy, so the separate endpoint maxima give a safe upper bound even when they do not occur at a common displacement.

In integer coordinates (r_l, r_u, a_l, a_u) the energy coordinate is bisected when $(r_u - r_l)K_0 \geq 2(a_u - a_l)r_u$, unless only the other coordinate remains available. The accepted grid areas sum exactly to $12G_0(K_0/2) = 1610612736$.

The explicit two-scale comparison. For $n = \lfloor z \rfloor$, subtraction gives

$$\mathcal{K}(z, x) - h_0(x) = \frac{\pi}{4}z + \frac{11\pi}{48}x - \left(1 - \frac{x}{12z}\right) \frac{R(z)}{z} - \frac{n}{12z} - 2\sqrt{1 - x^{-2}}. \quad (\text{B.5})$$

The quotient $R(z)/z$ is a sum of terms $\sqrt{1 - j^2/z^2}$, each increasing in z after activation. The factor $1 - x/(12z)$ increases in z and decreases in x . Therefore, on $[a, b] \times [c, d]$ within one fixed floor band, it suffices to check

$$P\left(\frac{c}{4} + \frac{11a}{48}\right) \geq \left(1 - \frac{a}{12d}\right) \text{upper} \frac{R(d)}{d} + \frac{n}{12c} + 2\frac{U(1 - b^{-2})}{B_0}. \quad (\text{B.6})$$

Here the normalized Riesz sum is enclosed directly by summing $U(1 - j^2/d^2)/B_0$.

Use denominator $G_2 = 2^{16}$ for both coordinates and initial rectangles $[1, 2] \times [n, n+1)$, $n = 1, 2, 3$. Intersect each box with $z \geq x$. A box with zero intersection area is discarded; all others are accepted by (B.6) or subdivided along a longest available coordinate, choosing x at a tie. The acceptance inequality proves the continuous frozen- n expression on the full closed box and hence on its clipped part within the half-open floor band. It does not substitute the old floor value at the next integer.

For a grid rectangle $[x_l, x_u] \times [z_l, z_u]$, twice its clipped grid area is computed exactly as

$$2(z_u - z_l) \max\{0, \min(x_u, z_l) - x_l\} + (v - u)(2z_u - v - u)$$

when $v > u$, with the last term omitted otherwise, where $u = \max(x_l, z_l)$ and $v = \min(x_u, z_u)$. The sums of accepted doubled areas on the three bands are $G_2^2, 2G_2^2, 2G_2^2$, exactly the corresponding initial clipped areas.

Within each fixed floor band, diagonal points $z = x$ are limits of the positive-area region above the diagonal. The finite collection of accepted closed boxes therefore covers them by continuity of the frozen- n expression. At $z = 1, 2, 3$ the band beginning there supplies its actual floor value; $z = 4$ belongs to (11.8). Shared subdivision edges are included. Thus the area checks are supplemented by a boundary argument and are not being used alone to infer coverage of spectral jumps.

Replay totals. The deterministic results are as follows.

Finite comparison	Tested	Accepted	Depth
Midpoint, $1/2 \leq y \leq 3$	245	123	12
$\mathcal{L} \geq 7/25, 1 \leq z < 7$	434	220	9
$\mathcal{L} \geq 8/25, 4 \leq z < 21$	1995	1006	10
$\mathcal{H} \leq 4/15, 2 \leq x \leq 4$	5395	2698	18
Center comparison, $1 \leq x \leq 2, x \leq z < 4$	9280	4620	18

All 8667 accepted boxes have positive rational margins. The triangle routine additionally skips 43 empty boxes. The program checks these statistics and the interval-width and grid-area identities against their exact expected values. The least margins, in table order, are

$$\overline{501827}, \quad \overline{39188871}, \quad \overline{3232271}, \quad \overline{39331}, \quad \overline{440730641},$$

$$\overline{4445962240}, \quad \overline{22229811200}, \quad \overline{4445962240}, \quad \overline{6668943360}, \quad \overline{125696244449280}.$$

These are enclosures on full intervals or boxes, not values sampled at selected parameters. All unbounded ranges were already reduced to the displayed finite regions by the analytic estimates in Section 11.

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NOTE ON AI USE

The author declares the use of consumer-grade Gemini-3 and GPT-6 models for prior-art search, exploring and developing ideas and proofs, coding, drafting and LaTeX typesetting; the author takes full and sole responsibility for all contents of this work.

REFERENCES

- [1] G. Pólya, On the eigenvalues of vibrating membranes, *Proc. London Math. Soc.* (3) **11** (1961), 419–433. doi:10.1112/plms/s3-11.1.419.
- [2] F. A. Berezin, Covariant and contravariant symbols of operators, *Math. USSR-Izv.* **6** (1972), no. 5, 1117–1151. doi:10.1070/IM1972v006n05ABEH001913.
- [3] P. Li and S.-T. Yau, On the Schrödinger equation and the eigenvalue problem, *Comm. Math. Phys.* **88** (1983), no. 3, 309–318. doi:10.1007/BF01213210.
- [4] A. Laptev, Dirichlet and Neumann eigenvalue problems on domains in Euclidean spaces, *J. Funct. Anal.* **151** (1997), no. 2, 531–545. doi:10.1006/jfan.1997.3155.
- [5] P. Freitas, J. Lagacé, and J. Payette, Optimal unions of scaled copies of domains and Pólya's conjecture, *Ark. Mat.* **59** (2021), no. 1, 11–51. doi:10.4310/ARKIV.2021.v59.n1.a2.
- [6] P. Freitas and I. Salavessa, Families of non-tiling domains satisfying Pólya's conjecture, *J. Math. Phys.* **64** (2023), no. 12, 121503. doi:10.1063/5.0161050.
- [7] X. He and Z. Wang, Pólya's conjecture for thin products, *Int. Math. Res. Not. IMRN* (2026), no. 16, rnag182. doi:10.1093/imrn/rnag182; arXiv:2402.12093v3.
- [8] N. Filonov, M. Levitin, I. Polterovich, and D. A. Sher, Pólya's conjecture for Euclidean balls, *Invent. Math.* **234** (2023), no. 1, 129–169. doi:10.1007/s00222-023-01198-1.
- [9] N. Filonov, M. Levitin, I. Polterovich, and D. A. Sher, Pólya's conjecture for Dirichlet eigenvalues of annuli, *J. London Math. Soc.* **113** (2026), no. 2, e70425. doi:10.1112/jlms.70425.
- [10] R. Jiang and F. Lin, Pólya's conjecture up to ϵ -loss and quantitative estimates for the remainder of Weyl's law, *Comm. Pure Appl. Math.* (2026), e70058, early view. doi:10.1002/cpa.70058; arXiv:2507.04307v4.
- [11] H. Weyl, Das asymptotische Verteilungsgesetz der Eigenwerte linearer partieller Differentialgleichungen (mit einer Anwendung auf die Theorie der Hohlraumstrahlung), *Math. Ann.* **71** (1912), no. 4, 441–479. doi:10.1007/BF01456804.
- [12] R. L. Frank and S. Larson, Riesz means asymptotics for Dirichlet and Neumann Laplacians on Lipschitz domains, *Invent. Math.* **241** (2025), 999–1079. doi:10.1007/s00222-025-01352-x.
- [13] A. Henrot, *Extremum Problems for Eigenvalues of Elliptic Operators*, Frontiers in Mathematics, Birkhäuser, Basel, 2006. doi:10.1007/3-7643-7706-2.
- [14] D. E. Horsley, Bessel phase functions: calculation and application, *Numer. Math.* **136** (2017), no. 3, 679–702. doi:10.1007/s00211-016-0853-7.
- [15] NIST Digital Library of Mathematical Functions, release 1.2.8 of 2026-09-15, <https://dlmf.nist.gov/>. See §§10.7, 10.9, 10.17, and 10.18, in particular formulas 10.9.4, 10.17.3, and 10.17.9.
- [16] Github Repository - [www.github.com/aron-chandru/arbitrary_square](https://github.com/aron-chandru/arbitrary_square).

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